Electronics and Communications Department
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SafeLink

Graduation Project is submitted to the Electronics and Communications Department in partial fulfilment of the requirements for the Degree of Bachelor of Science, Electronics and Communications Engineering

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SafeLink

A Wearable Proximity Safety System

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Team SafeLink

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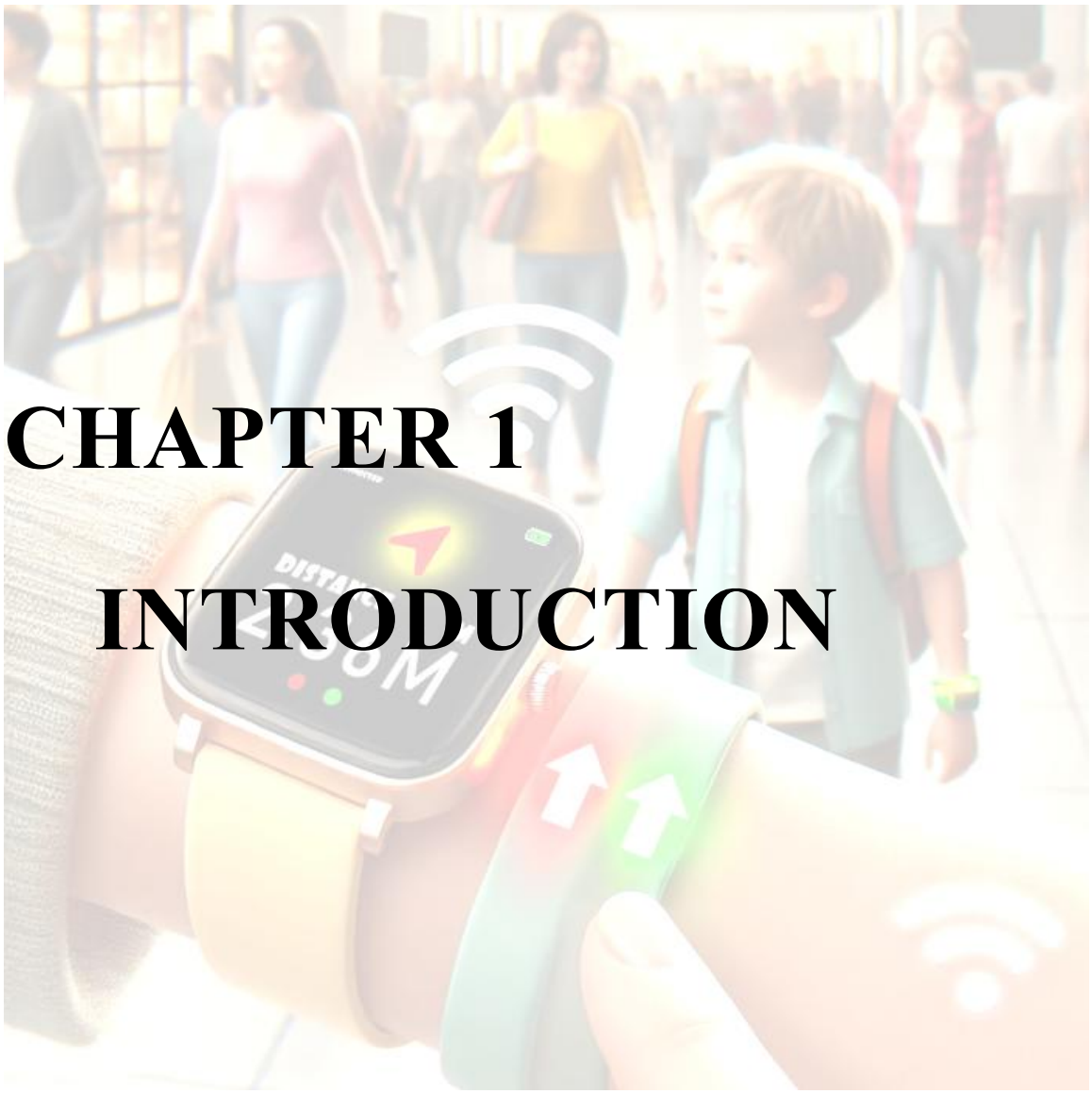
Abstract

This project introduces a smart wearable system aimed at enhancing safety, awareness, and monitoring for two particularly vulnerable groups in society: children susceptible to abduction and elderly individuals living with dementia or Alzheimer's disease. These groups face daily risks that may compromise their well-being, especially in crowded public areas or unsupervised environments. The system is built around the concept of maintaining a continuous and intelligent link between a caregiver and the person under their supervision, ensuring prompt alerts and intuitive guidance when potential danger is detected.

The primary goal of the project is to provide a reliable, real-time solution that can help caregivers respond quickly if a child wanders away or if an elderly individual strays beyond a safe distance. By enabling continuous awareness of separation events and directional information, the system contributes significantly to minimizing the time needed to respond and reestablish proximity. Additionally, the project is designed to preserve freedom of movement for both children and elderly individuals while still ensuring the safety that comes with consistent monitoring.

This solution proposes a balance between technological innovation and human caregiving, offering an effective alternative to traditional surveillance or GPS-based systems. It promotes autonomy and mobility while reducing the anxiety and stress often experienced by parents and caregivers. The project focuses on real-world applicability by emphasizing usability, affordability, and responsiveness.

Through this work, we aim to bridge a critical gap in personal safety solutions—one that supports early intervention without relying on invasive tracking or complex interfaces. The system stands as a testament to how wearable technology can be purposefully and ethically applied to make everyday life safer, more manageable, and more secure for those who need it most.



CHAPTER 1

INTRODUCTION

CHAPTER 1: Introduction

1.1 Overview of the Project

1.1.1 Child-Elder Safety in Modern Environments

In today's rapidly evolving world, ensuring the safety of vulnerable individuals such as children and elderly people is more critical than ever. With rising urban congestion, crowded environments, and cognitive disorders like Alzheimer's affecting millions of seniors globally, caregivers face immense difficulty maintaining continuous awareness of their loved ones' location and safety.[1]

1.1.2 Importance of Real-time Proximity Awareness

Traditional GPS-based or cellular tracking solutions often fall short due to signal inaccuracy indoors, battery drain, and dependency on subscriptions or infrastructure. A wearable system that enables real-time separation alerts and direction guidance within a limited range can address this problem with simplicity and reliability.[4][5]

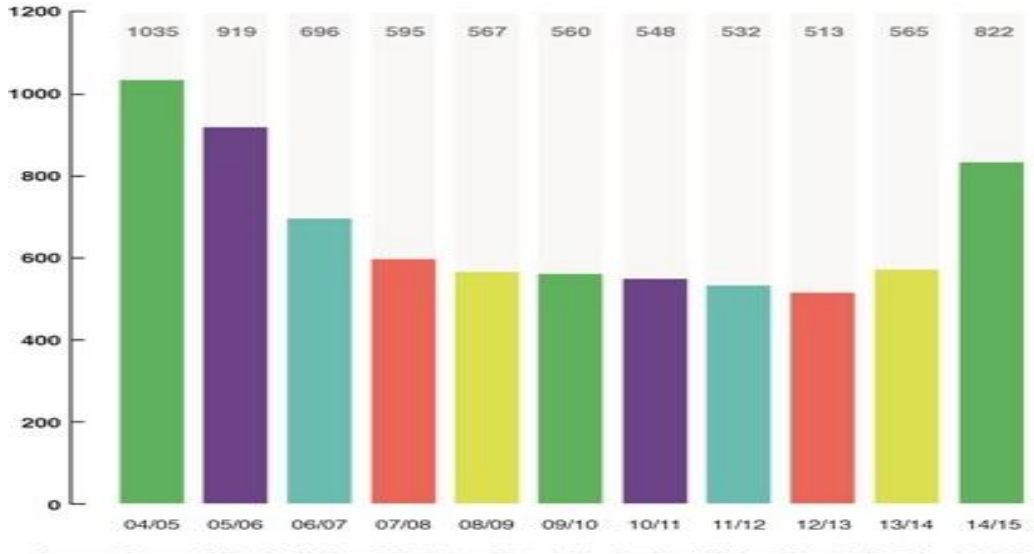


Figure 1-1:Number of Police recorded offenses of child abduction in England

1.2 Problem Statement

Many safety systems lack short-range directional feedback and require expensive infrastructure, making them inaccessible or unreliable in certain regions. Caregivers need a non-intrusive, localized method for detecting if a dependent individual strays too far and determining how to quickly reunite.

1.3 Objectives

1.3.1 Primary Goals

- Provide a wearable safety solution to detect separation between two users.
- Offer directional and distance-based guidance to reconnect caregivers and dependents.
- Trigger real-time alerts through vibration, sound, and visual feedback.

1.3.2 Secondary Goals

- Ensure the device operates without GPS or mobile networks.
- Maximize battery life and user comfort.
- Enable system scalability for institutional use (schools, hospitals).

1.4 Target Users and Applications

1.4.1 Parent-Child Use Case

Parents can use the system in parks, malls, and public transport to prevent child separation.

1.4.2 Elderly Care Use Case

Caregivers of seniors with dementia can rely on proximity alerts if the elderly individual moves out of a defined safe zone.

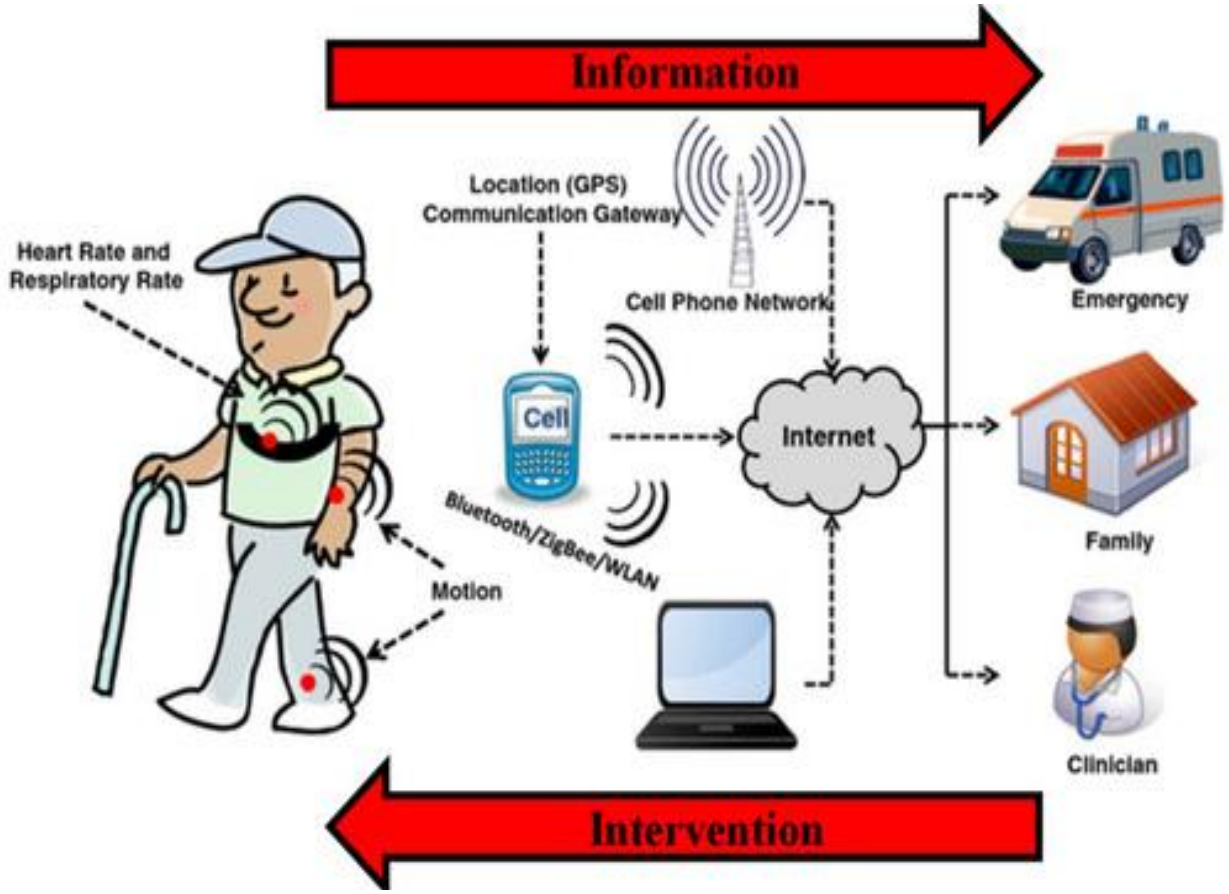


Figure 1-2:Elderly Caregiving

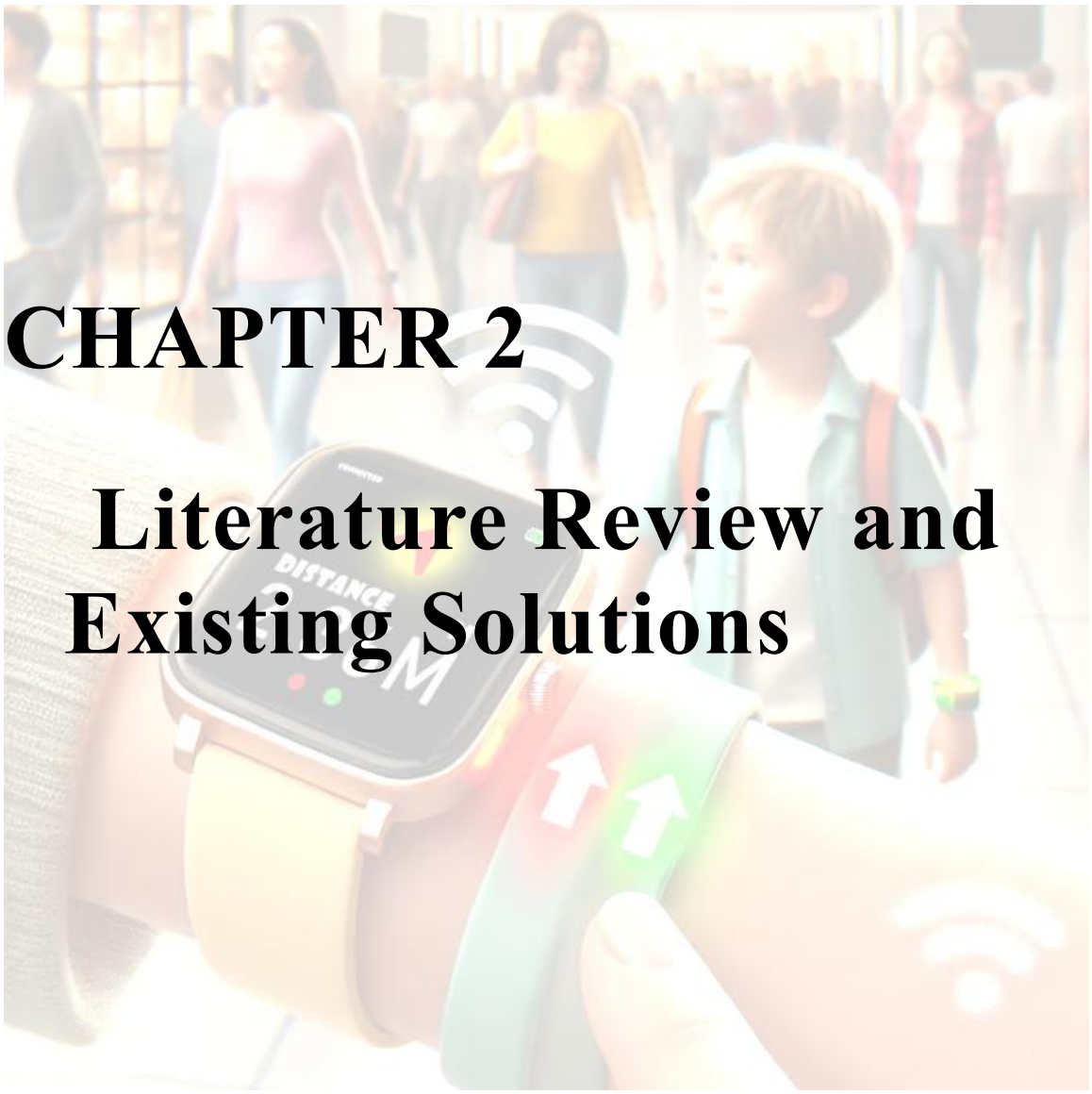
1.5 Scope and Limitations

Scope:

- Two wearable bracelets are linked via local wireless communication.
- Directional and alert feedback system.
- Compass-based heading detection and RSSI-based proximity estimation.

Limitations:

- Limited range due to Wi-Fi/BLE constraints.
- Compass sensitivity to metal and orientation.
- No GPS, so it is not suitable for outdoor navigation or location history.



CHAPTER 2

Literature Review and Existing Solutions

CHAPTER 2: Literature Review and Existing Solutions

2.1 Market-Available Trackers

2.1.1 GPS-Based Solutions

Global Positioning System (GPS) trackers are among the most common solutions for monitoring the location of children and elderly individuals. Devices such as Jiobit, AngelSense, and Verizon GizmoWatch utilize GPS satellites and cellular networks to send real-time location updates to a paired smartphone. These solutions offer wide coverage but suffer from limitations like poor indoor accuracy, short battery life, and dependency on paid subscriptions.

Figure 2-1:Jiobit

Figure 2-2:Angel Sense

2.1.2 Bluetooth Low Energy Trackers

BLE-based devices like Tile or Apple AirTag provide short-range tracking using signal proximity to a smartphone. These trackers are energy-efficient and affordable, but their detection range is typically limited to 10–30 meters, and they lack directional guidance.

Figure 2-3:BLE-Tile

Figure 2-4:Apple AirTag

2.1.3 LoRa-based Trackers

LoRa (Long Range) devices use low-power, wide-area networks (LPWAN) to enable location reporting over kilometers. While promising, these solutions require dedicated gateways, and public LoRa infrastructure is sparse in many regions, including Egypt. Additionally, LoRa is not ideal for indoor environments or short-range directional detection.

Figure 2-5:LoRa Trackers

2.2 Academic and Open-Source Projects

Several academic projects and open-source initiatives have explored alternatives to GPS and cellular tracking. Examples include indoor navigation using Wi-Fi RSSI, compass-assisted direction estimation, and wearable alert systems. These approaches often demonstrate innovation but are typically tested only in controlled environments and lack robustness for real-world deployment.

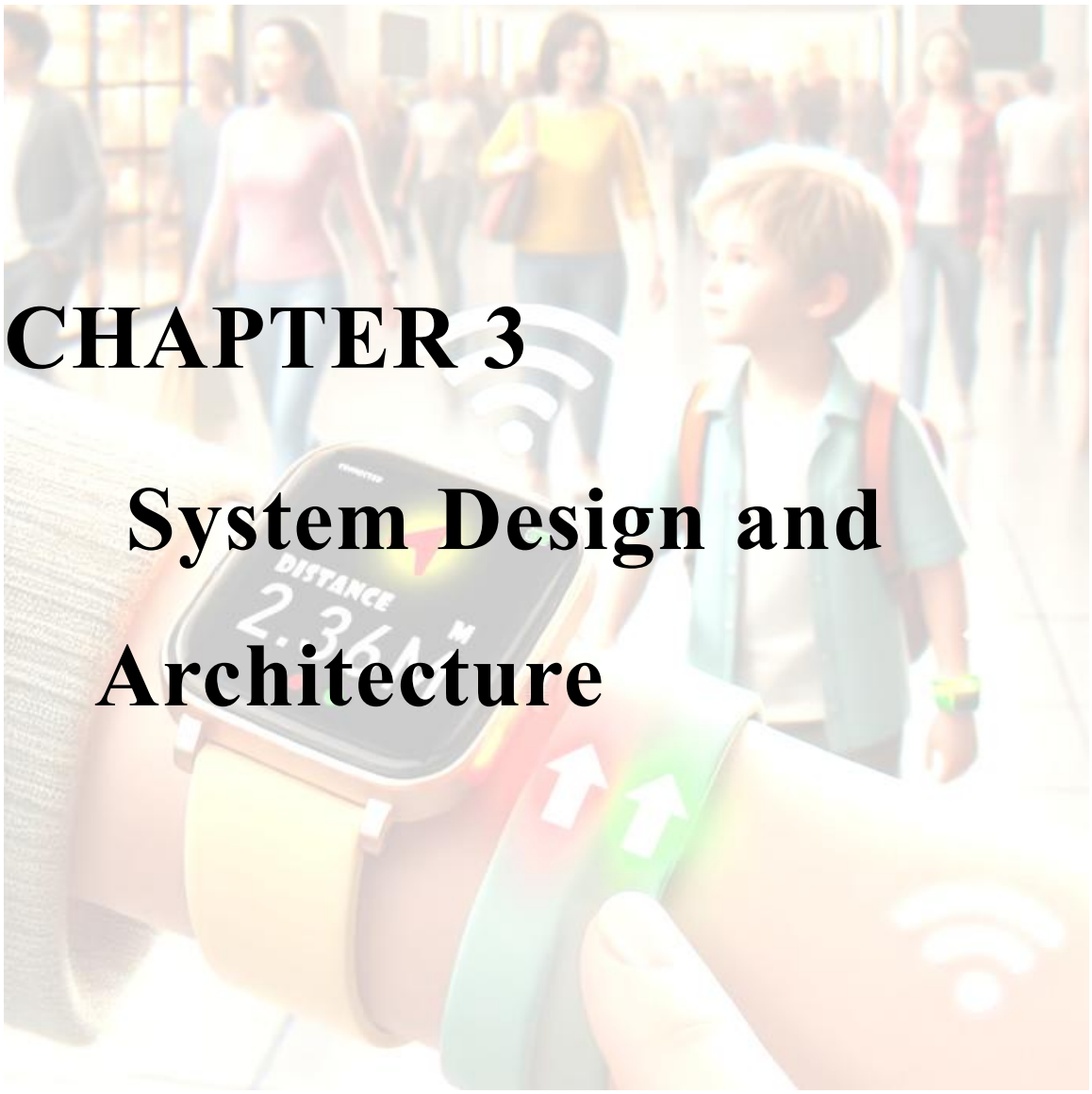
2.3 Comparative Analysis

Commercial and academic solutions each offer unique advantages and trade-offs:

- GPS: Long-range but costly and energy intensive.
- BLE: Affordable and power-efficient but short-range.
- LoRa: Long-range with minimal power use but requires infrastructure.
- Wi-Fi RSSI: Medium-range and infrastructure-light, making it ideal for SafeLink.

2.4 Technological Gaps and Innovation Points

Existing systems often neglect the need for immediate, local, real-time feedback when separation occurs. Moreover, few provide directional guidance based on the dependent's relative position. SafeLink addresses these gaps with RSSI-based distance estimation and multiple feedback mechanisms in a low-cost wearable system.[14]



CHAPTER 3

System Design and Architecture

CHAPTER 3: System Design and Architecture

3.1 Functional Description

3.1.1 Parent Bracelet

The parent bracelet is the central monitoring unit of the system. It receives wireless data from the child bracelet and provides real-time feedback through a graphical display, vibration motor, and buzzer. It features a directional arrow indicating the relative heading to the child, a distance estimate, and battery indicators. The parent device is wearable, rechargeable, and designed to be as unobtrusive as possible while remaining responsive

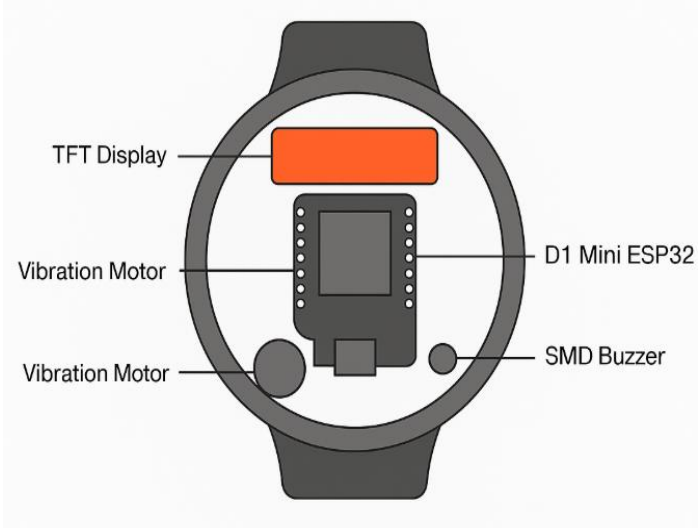


Figure 3-1:Parent Bracelet Design Prototype

3.1.2 Child Bracelet

The child bracelet functions as the tracked unit. It continuously transmits signal strength and orientation-related data to the parent bracelet. It includes tamper detection via a reed switch, low-battery monitoring, and a compact, child-safe enclosure. The child device is also wearable and powered by a rechargeable battery.

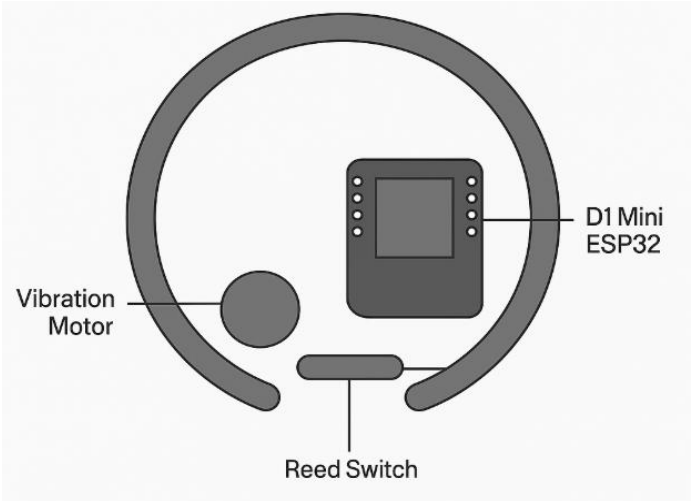


Figure 3-2:Child Bracelet Design Prototype

3.2 Hardware Architecture

Both bracelets are built around the D1 Mini ESP32 microcontroller, which supports dual Wi-Fi modes and low-power operations. The parent bracelet includes a circular TFT display, compass sensor, buzzer, vibration motor, and voltage divider for battery sensing. The child bracelet includes similar communication capabilities and sensors but omits the display.

The hardware is modular and designed to be integrated into a custom PCB for compactness and reliability. Each bracelet operates on a 3.7V Li-Po battery managed through TP4056 charging modules.

3.3 Software Architecture

The software architecture is based on firmware developed using Arduino IDE and structured in modules:

- Communication: Wi-Fi setup, packet structure, reconnection logic.
- Data Processing: RSSI averaging, compass vector processing.
- Display Module: Graphical arrow, text display, color-coded alerts.
- Alert Module: Triggers vibration, buzzer, and display messages.

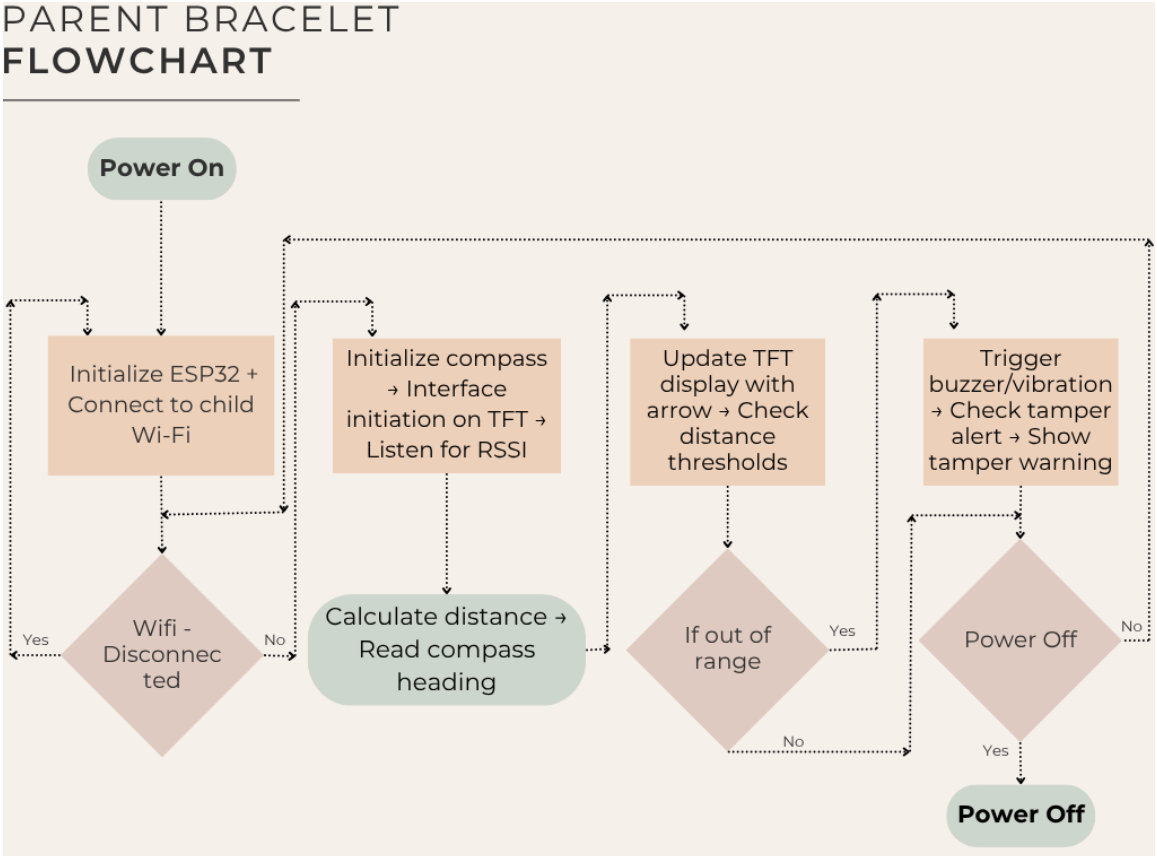


Figure 3-3:Parent Bracelet Flowchart

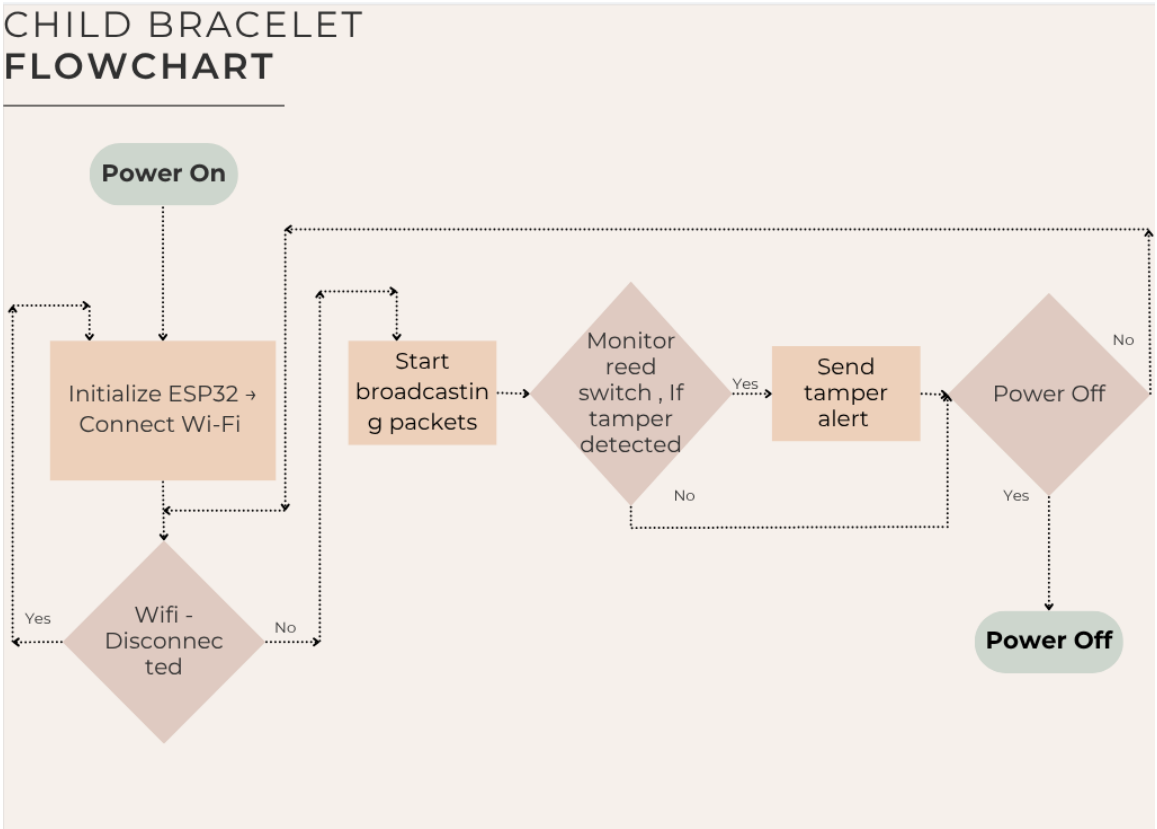


Figure 3-4:Child Bracelet Flowchart

3.4 Data Flow Diagram

The system follows unidirectional data flow from the child to the parent device. The child broadcasts data packets containing RSSI. The parent device receives and processes these packets to determine proximity and heading, then activates the appropriate feedback mechanisms.[19][20]

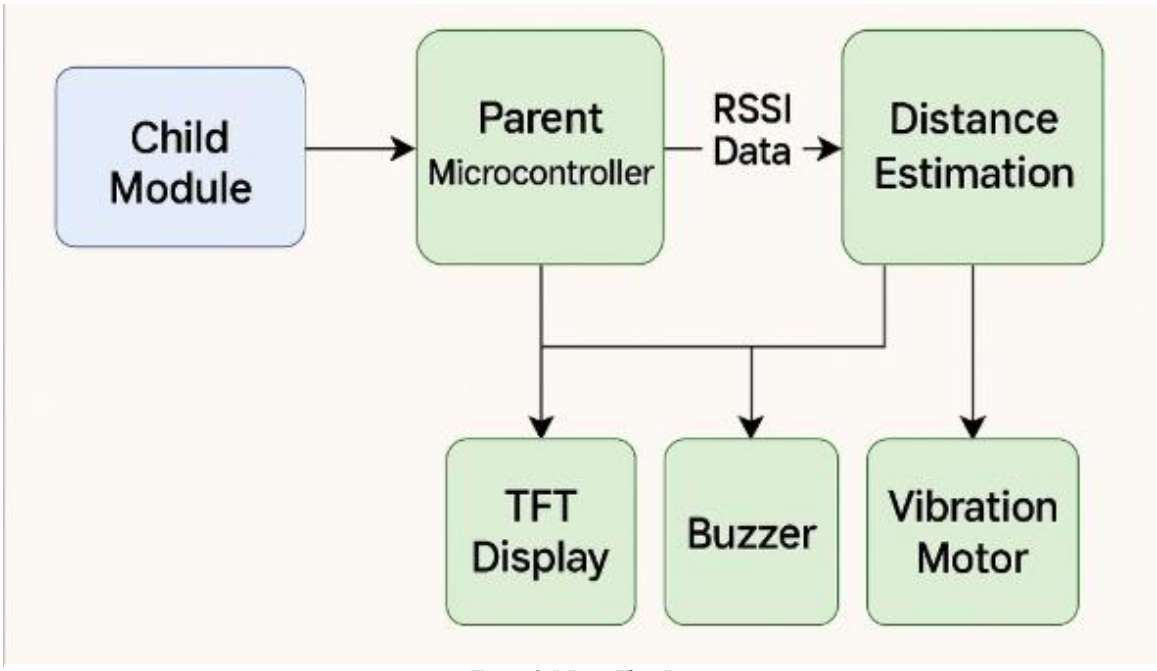


Figure 3-5:Data Flow Diagram

3.5 Block Diagram and System Workflow

The system workflow can be summarized as follows:

1. Child device continuously sends beacon packets.
2. Parent receives data and calculates RSSI and compass vector.
3. Distance and direction are displayed.
4. If separation exceeds a threshold, alerts are triggered.

3.6 Power Supply Management

Each bracelet is powered by a 3.7V Li-Po battery. The TP4056 charging module ensures safe USB charging. Voltage dividers feed battery status to the ESP32 ADC. The firmware is optimized to minimize power consumption by using sleep modes during inactivity and regulating display brightness and Wi-Fi transmission intervals.

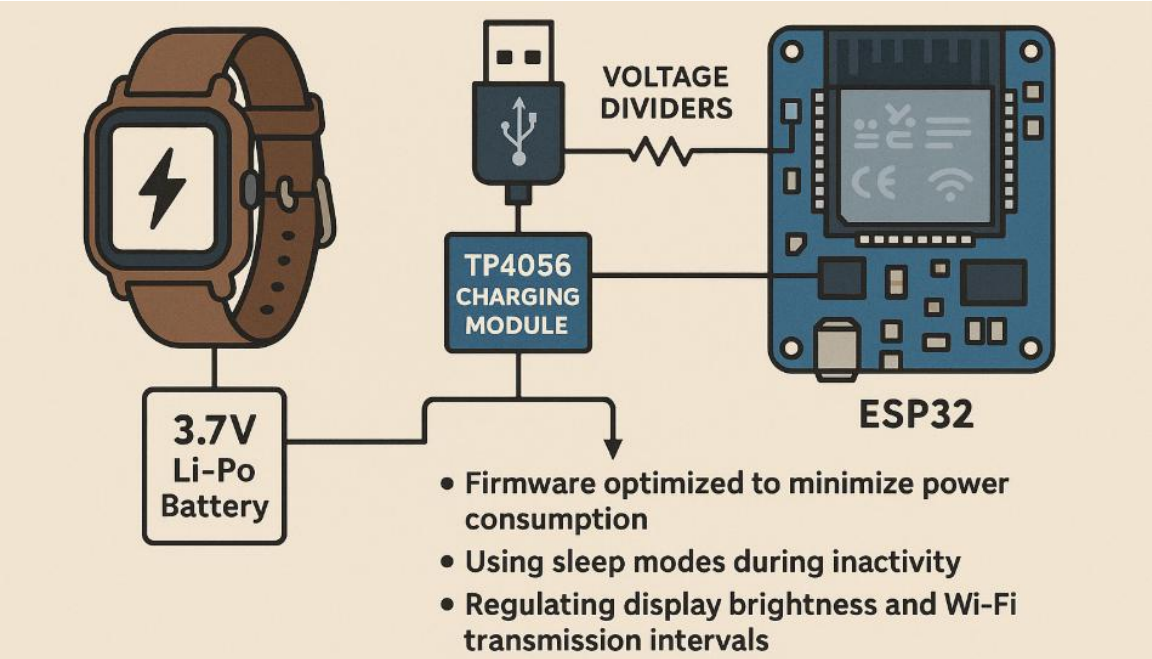
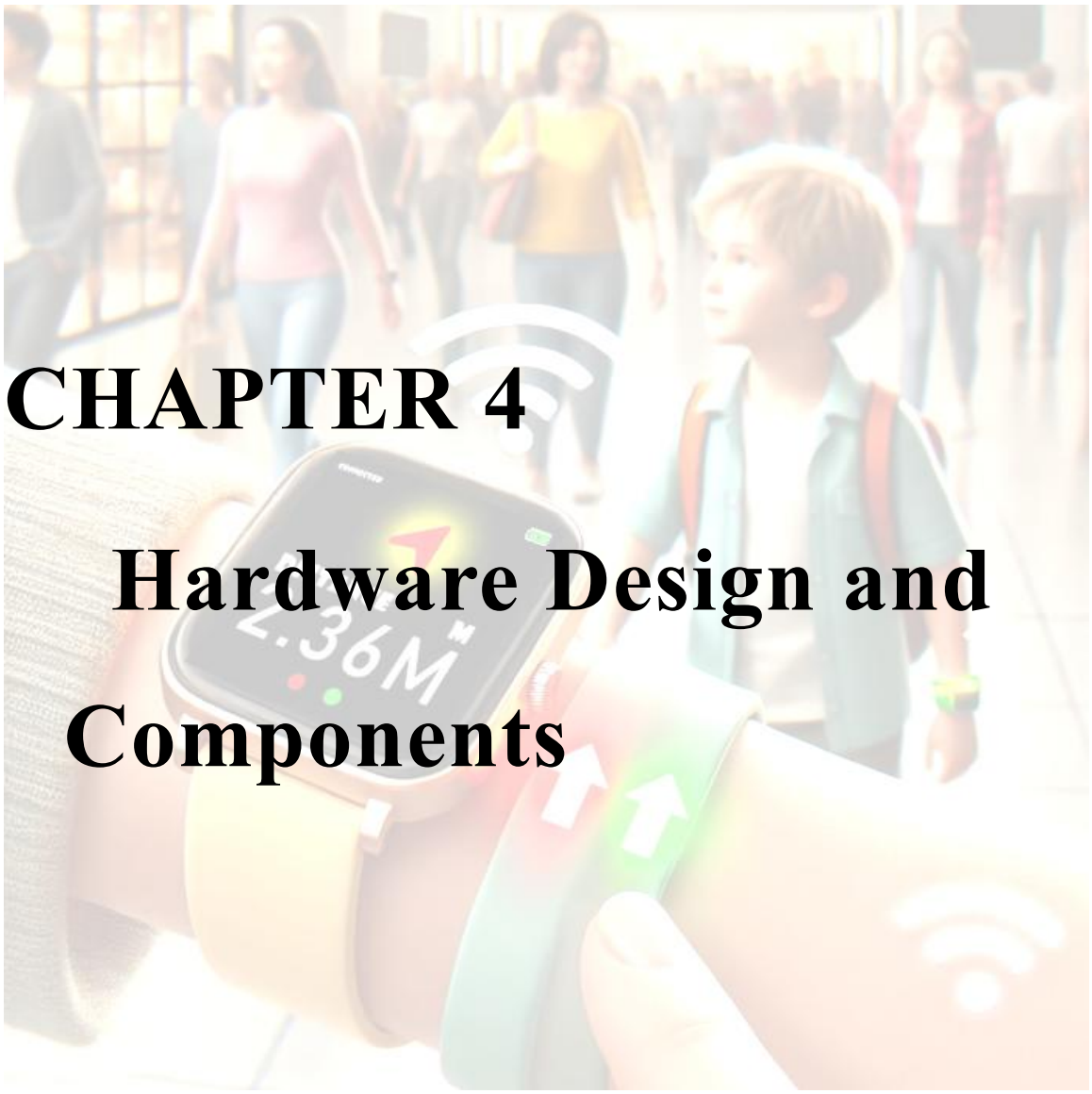


Figure 3-6:Power Supply Management



CHAPTER 4

Hardware Design and Components

CHAPTER 4: Hardware Design and Components

4.1 Microcontroller Unit

D1 Mini ESP32 The core of both bracelets is the D1 Mini ESP32, a compact development board that integrates Wi-Fi and Bluetooth functionality. It offers dual-core processing, low-power sleep modes, and sufficient I/O pins for handling displays, sensors, and output devices. Its small form factor makes it ideal for wearable applications.

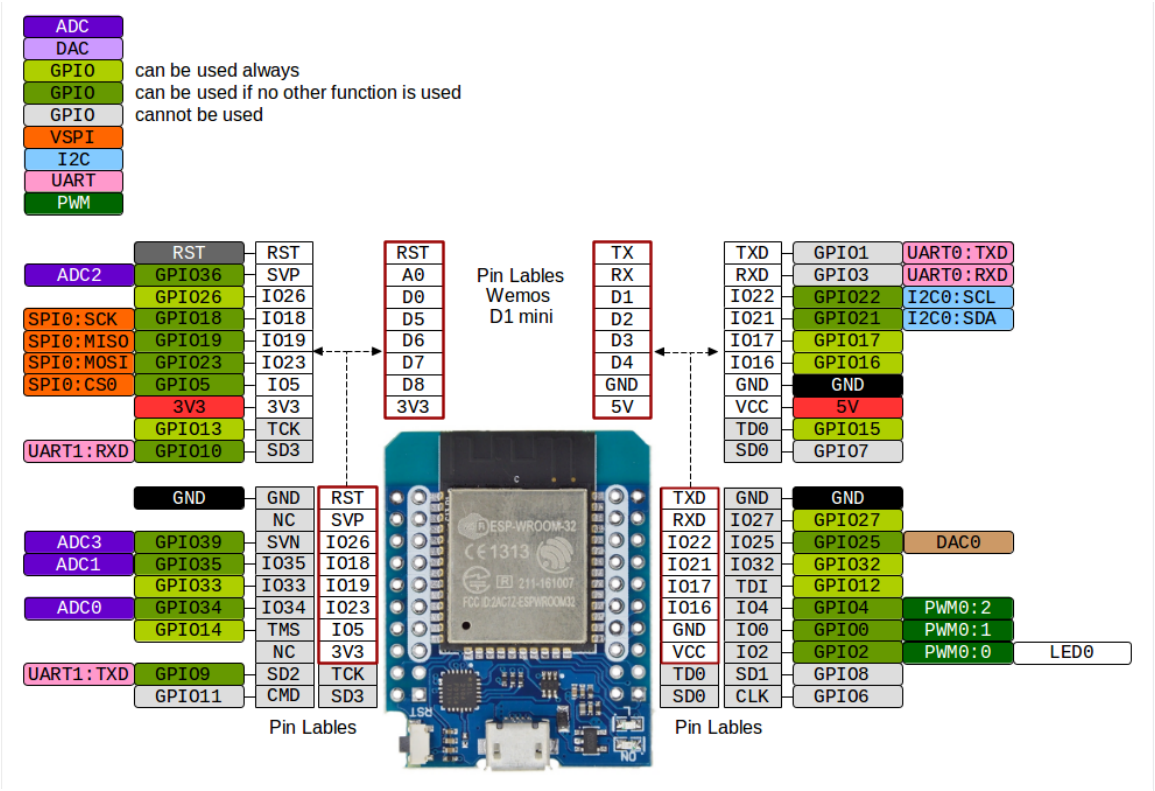


Figure 4-1:pinout diagram of D1 Mini ESP32

4.2 Compass Sensor (QMC5883L)

The QMC5883L is a 3-axis digital compass sensor used in the parent bracelet to determine heading direction. It outputs magnetometer data over I2C and is used in calculating the relative direction from parent to child. It requires periodic calibration to maintain accuracy.

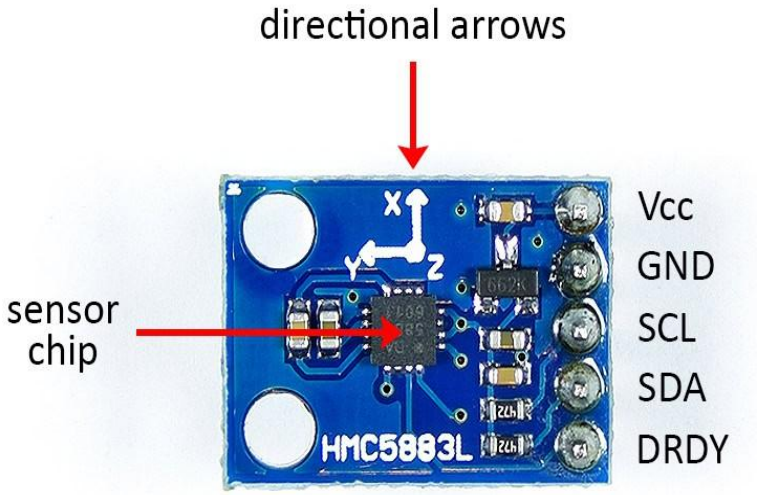


Figure 4-2:QMC5883L sensor pinout

4.3 Display (GC9A01 TFT screen)

The parent bracelet includes a 1.28-inch circular TFT LCD based on the GC9A01 driver. This screen displays an arrow pointing toward the child, as well as textual alerts and battery levels. The compact and high-contrast display enables quick interpretation of information by the user.

Figure 4-3:GC9A01 TFT screen pinout

4.4 Vibration Motor

A coin-type vibration motor is used in both bracelets for silent alerts. It activates during separation events or tamper detection, providing physical feedback to the wearer without drawing attention.

Figure 4-4:Coin vibration motor

4.5 Piezo Buzzer

The parent bracelet contains a KLJ-8530-3627 SMD piezo buzzer. It produces an audible tone when alerts are triggered, complementing the vibration feedback. Its small footprint and high sound output make it suitable for wearable use.

Figure 4-5:SMD piezo buzzer

4.6 Reed Switch

for Tamper Detection The child bracelet integrates a reed switch that detects if the bracelet is removed or tampered with. This magnetic-based component triggers alerts when the circuit is broken, indicating that the bracelet may have been taken off.

Figure 4-6:Reed Switch

4.7 Battery and Charging Module (TP4056)

Each bracelet is powered by a 3.7V Li-Po battery. The TP4056 module manages USB charging, providing overcharge and over-discharge protection. It also includes an onboard battery status LED indicator.

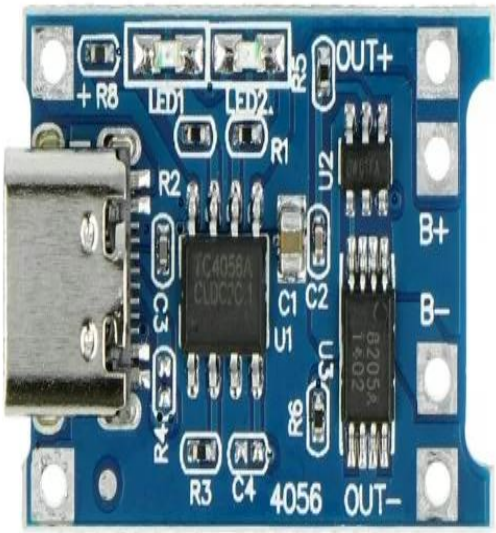
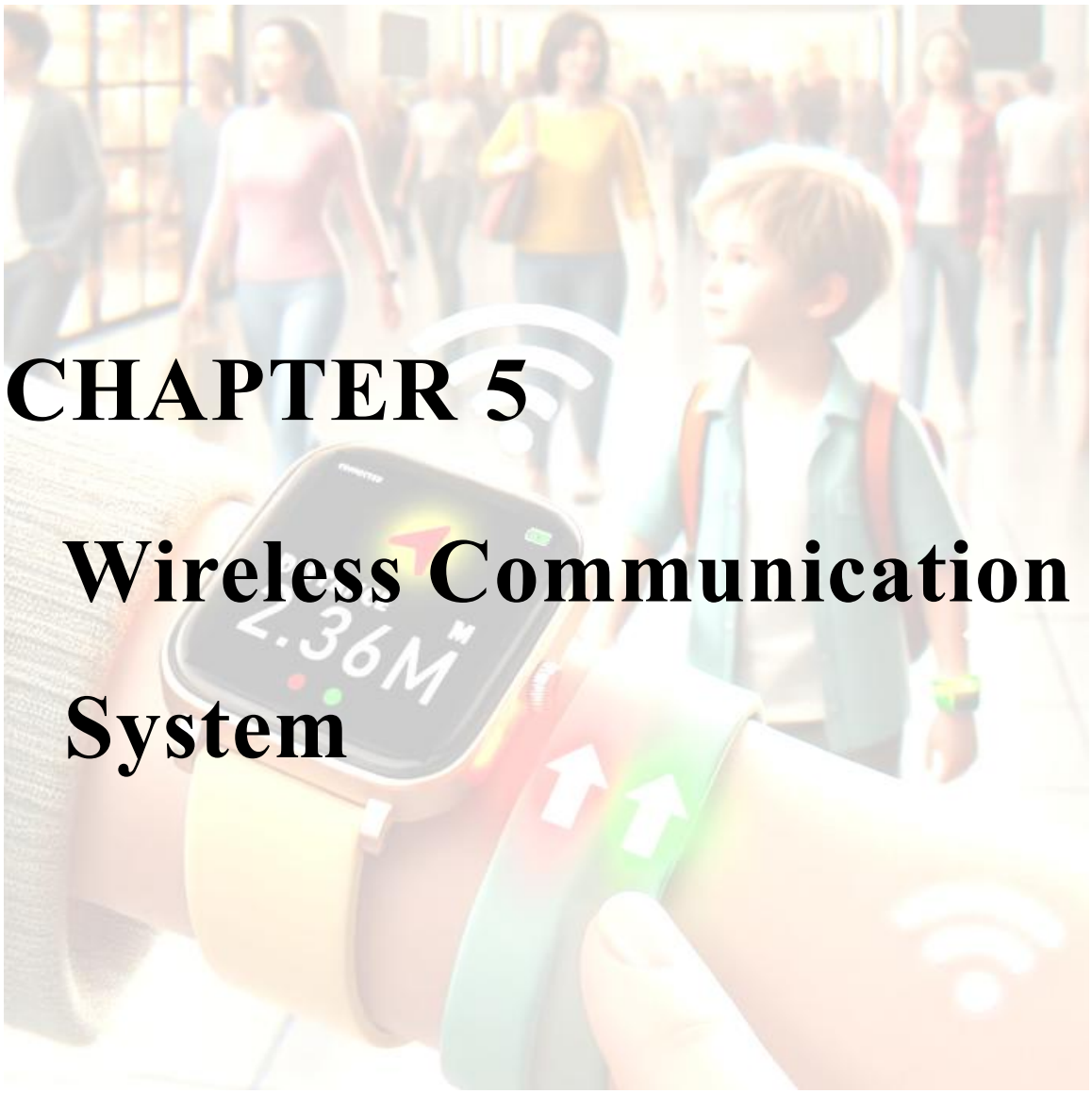


Figure 4-8:TP4056 charging module

Figure 4-7:Li-Po Battery

4.8 Voltage Divider for Battery Monitoring

To monitor battery voltage, a simple voltage divider circuit steps down the Li-Po voltage to within the ADC range of the ESP32. This allows the system to estimate and display battery percentage on the parent bracelet.



CHAPTER 5

Wireless Communication System

CHAPTER 5: Software Design and Implementation

5.1 Wi-Fi AP + STA Mode

The SafeLink system operates using Wi-Fi communication between the child and parent bracelets. The child bracelet is configured in Access Point (AP) mode, while the parent bracelet operates in Station (STA) mode. This allows the parent to connect directly to the child without relying on external routers or internet access. The AP+STA configuration is essential for establishing a closed and local wireless link that supports RSSI-based proximity estimation.

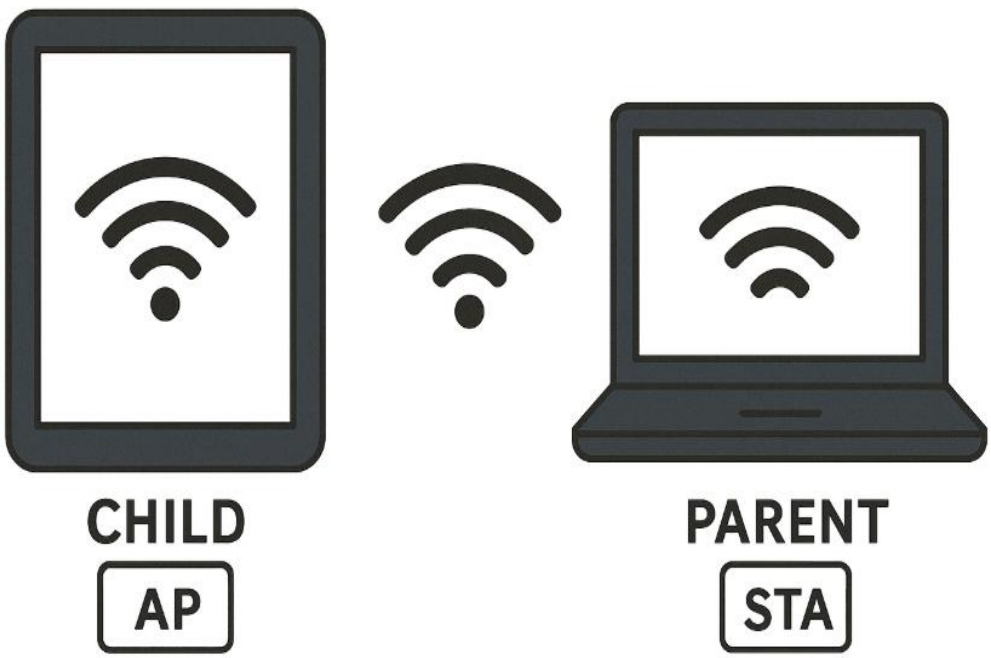


Figure 5-1:WiFi AP <-->STA Connection

5.2 RSSI-based Distance Estimation

5.2.1 RSSI Theory Received Signal Strength Indicator (RSSI)

RSSI is a measure of power present in a received radio signal. In wireless communication systems, the strength of a received signal generally decreases as the distance between the transmitter and receiver increases. SafeLink uses this relationship to estimate the proximity between child and parent devices, enabling directional and distance-based feedback.[2]

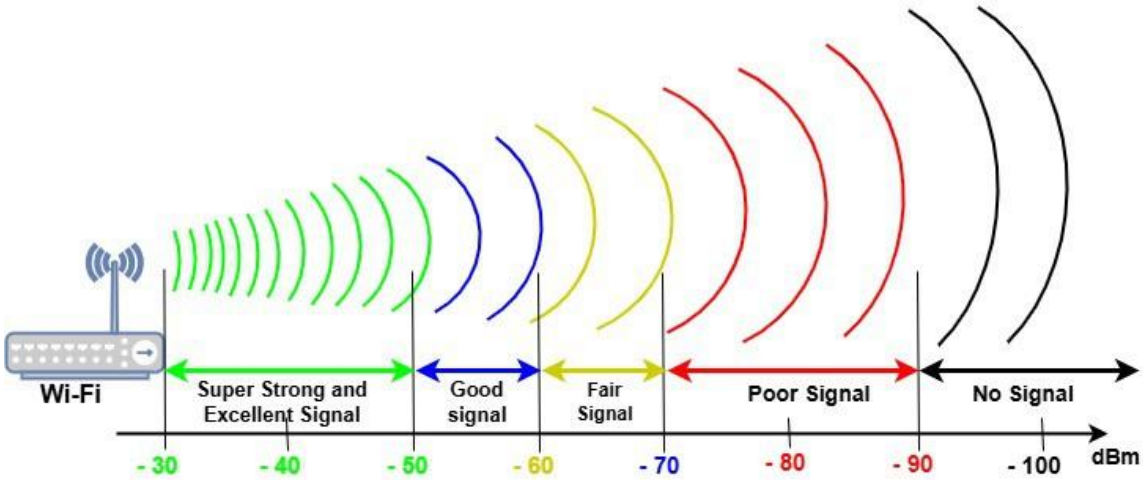


Figure 5-2: WiFi RSSI

5.2.2 Calibration and Environmental Impact

RSSI values can fluctuate due to interference, reflections, and obstacles. Therefore, a calibration routine was performed to map RSSI values to approximate distances in a typical usage environment. The system uses a sliding average to smooth signal noise and enhance stability. However, exact distance measurement is not guaranteed; the system classifies distance into three zones: close, moderate, and far.

5.3 Communication Protocol and Data Format

The child device continuously broadcasts data packets containing minimal payloads, such as a unique ID and signal timestamp. The parent device listens for these packets, extracts the RSSI value, and uses the data for proximity estimation. The packet structure is optimized to reduce latency and conserve power.

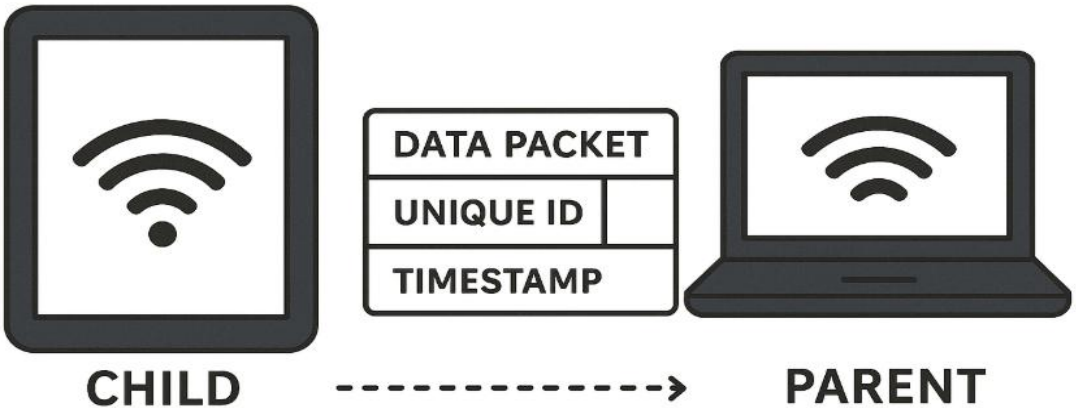


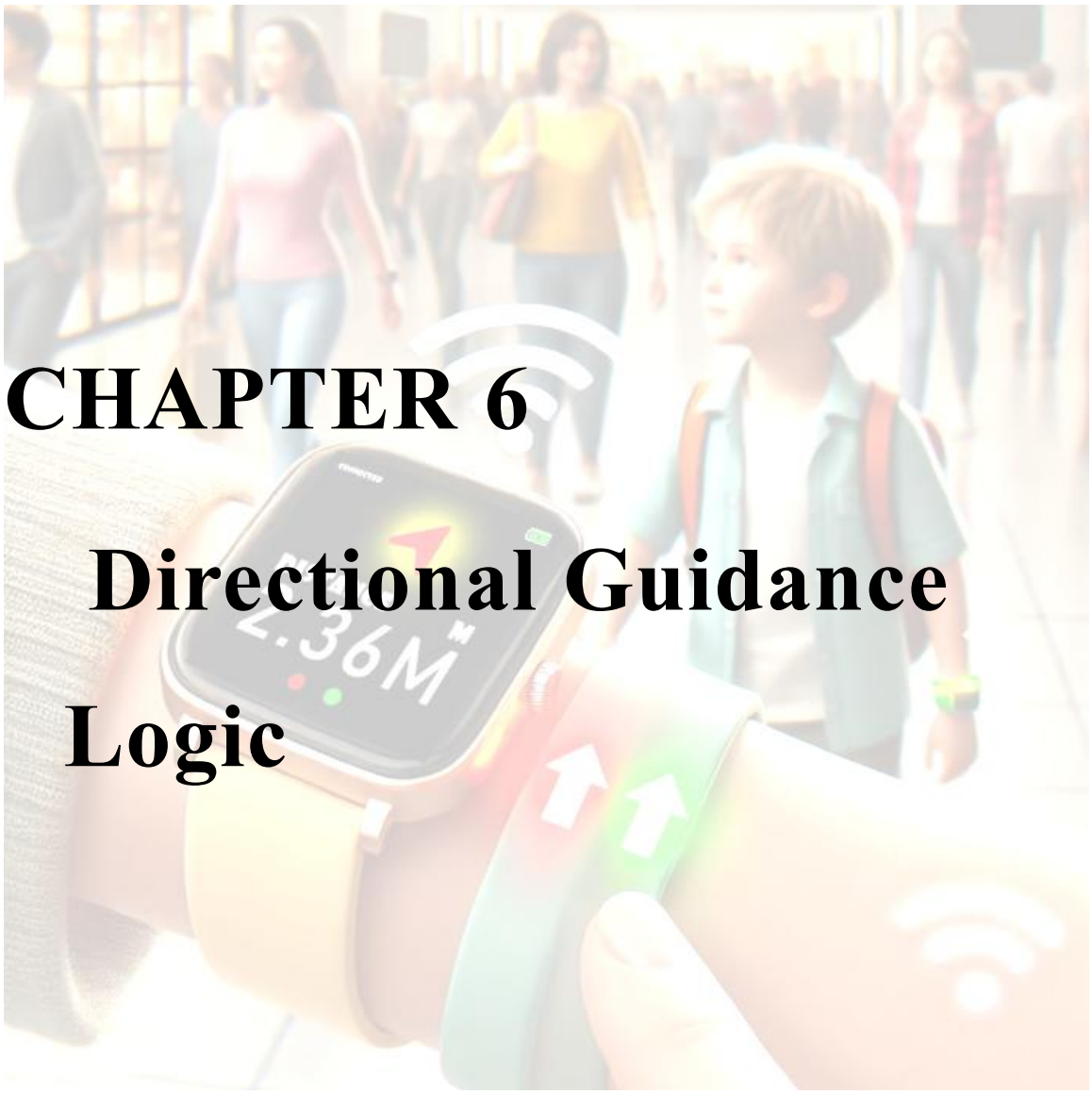
Figure 5-3:Data Communication Between the child and parent

5.4 Connection Management and Reconnection Logic

To ensure reliability, the parent device includes logic to automatically reconnect if the signal is lost. Timeouts are used to detect disconnection, and upon reconnection, the system resumes normal operation without requiring manual intervention. This enhances user experience and system robustness in dynamic environments.

5.5 Limitations of RSSI and Possible Enhancements

While RSSI provides a useful approximation for distance, it is affected by multi-path fading and environmental factors. Future improvements may include filtering algorithms (e.g., Kalman filter) or combining RSSI with Time-of-Flight or UWB modules for greater precision. Nonetheless, RSSI remains a practical choice for this wearable system due to its simplicity and low overhead.



CHAPTER 6

Directional Guidance Logic

CHAPTER 6: Directional Guidance Logic

6.1 Compass Sensor Operation

The SafeLink system uses the QMC5883L compass sensor on the parent bracelet to determine heading direction. This 3-axis magnetometer detects the Earth's magnetic field and outputs raw vector data. It is interfaced with the ESP32 via I2C and regularly polled to obtain real-time directional values. The heading is calculated relative to magnetic north and is used to align the display arrow toward the child's position.[24][25]

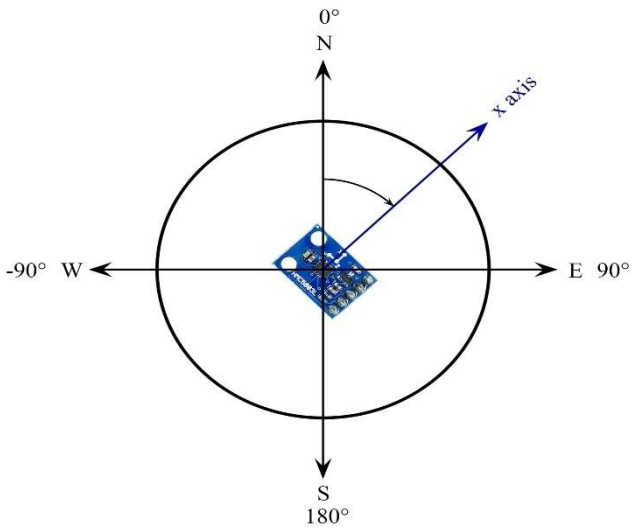


Figure 6-1:QMC5883L magnetometer working mechanism

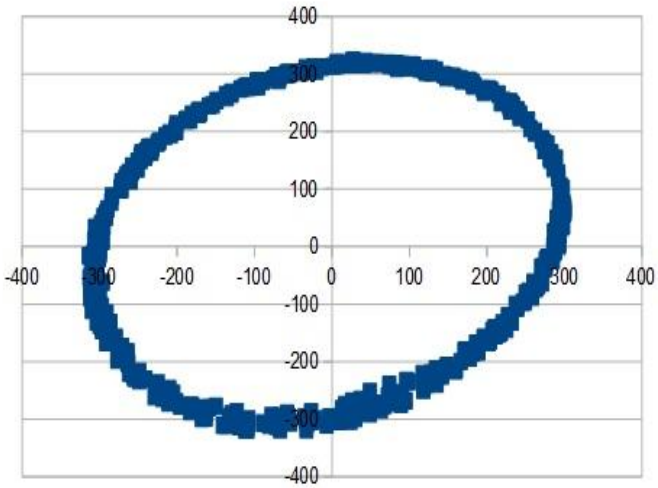


Figure 6-2QMC5883L calibration

6.2 Angle Calculation Methodology

6.2.1 Magnetometer Vector Processing

The sensor provides magnetic field values along the X and Y axes. These values are processed to compute the heading angle using the arctangent function: heading = atan2(Y, X)

The result is then normalized to a 0–360° scale. To improve accuracy, offsets due to hard-iron distortions are applied, and the readings are smoothed using moving averages.

6.2.2 Parent-Child Relative Heading

The RSSI signal from the child indicates relative proximity, but not direction. By combining the parent's orientation from the compass with a reference calibration (initial position during sync), a virtual direction vector is established. This helps infer whether the child is ahead, behind, or to the side.

6.3 Displaying Arrows on TFT

An arrow is rendered on the GC9A01 circular TFT display of the parent bracelet. Based on the calculated heading, the arrow rotates to point toward the child's direction. The display updates dynamically, with smooth visual transitions. This intuitive visualization helps the parent react quickly to reposition or approach the child.

6.4 Arrow Color Based on Distance

To enhance awareness, the arrow color changes according to the proximity zone derived from RSSI values:

6.4.1 Green

Close Indicates that the child is within a safe, close range. No alerts are triggered.

6.4.2 Yellow

Moderate Suggests increasing distance but still within communication range. A cautionary state.

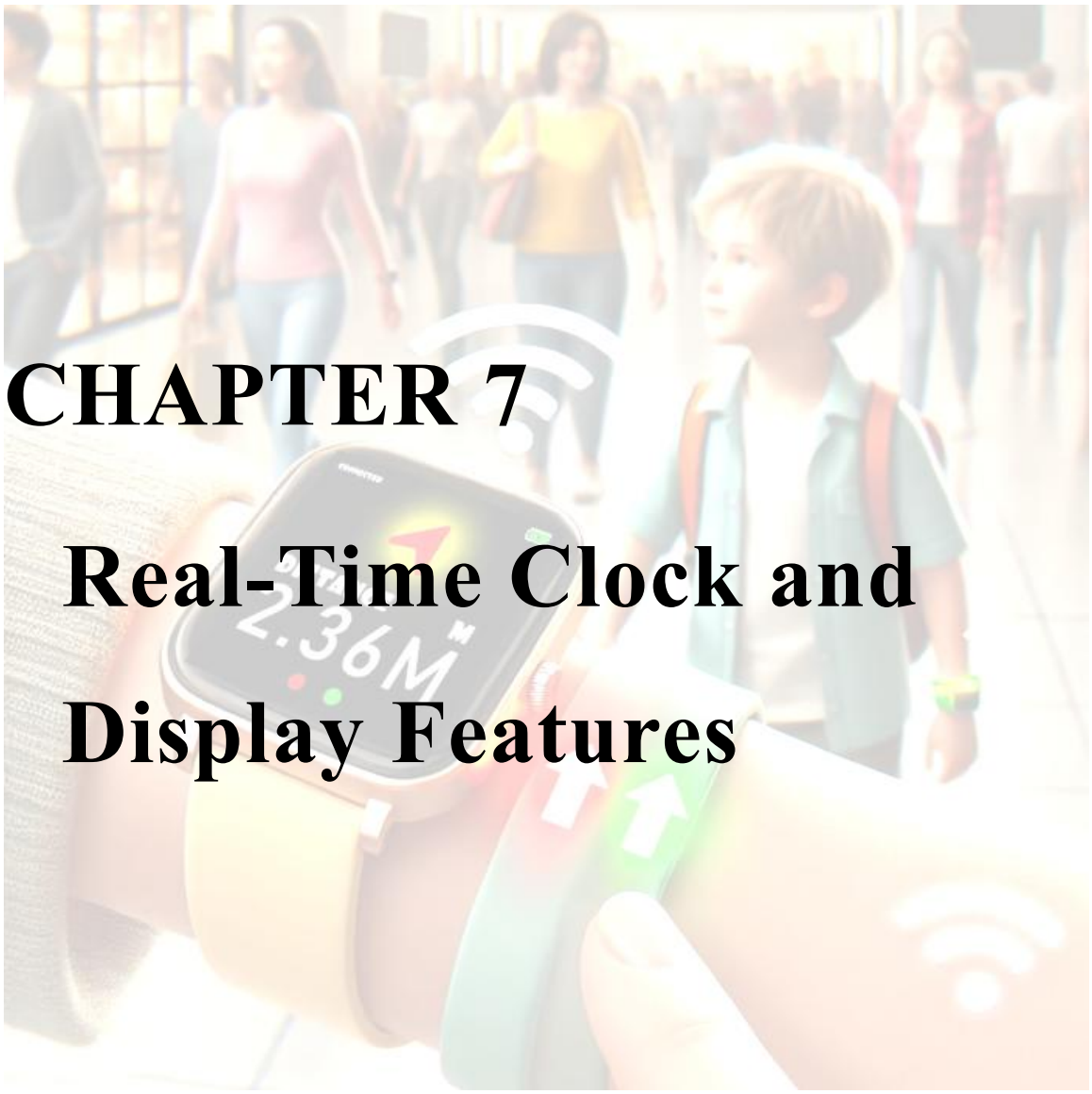
6.4.3 Red

Out of Range Triggers full alerts including vibration and buzzer. The child has likely moved too far from the parent.

6.5 Issues and Adjustments for Compass Accuracy

Compass readings are sensitive to magnetic interference, device tilt, and nearby metal objects. To address this, the system:

- Applies offsets obtained from calibration.
- Filters noisy data using moving averages.
- Recommends a short calibration spin during first-time use.
- Disables directional guidance if compass values become erratic or invalid.



CHAPTER 7

Real-Time Clock and Display Features

CHAPTER 7: Real-Time Clock and Display Features

7.1 NTP Time Synchronization

Accurate timekeeping is essential for real-time monitoring systems, particularly when the user interface includes timestamps or log data. In SafeLink, the ESP32's built-in Wi-Fi capabilities enable connection to Network Time Protocol (NTP) servers. During the initial setup or upon every reboot, the parent device connects to an available Wi-Fi network temporarily and retrieves the current Coordinated Universal Time (UTC). This value is then adjusted according to the local time zone and stored in system memory. Time synchronization occurs periodically in the background to maintain accuracy over extended use. This ensures that any alerts, system logs, or future time-based features (e.g., alarms or timed shutdowns) function reliably.

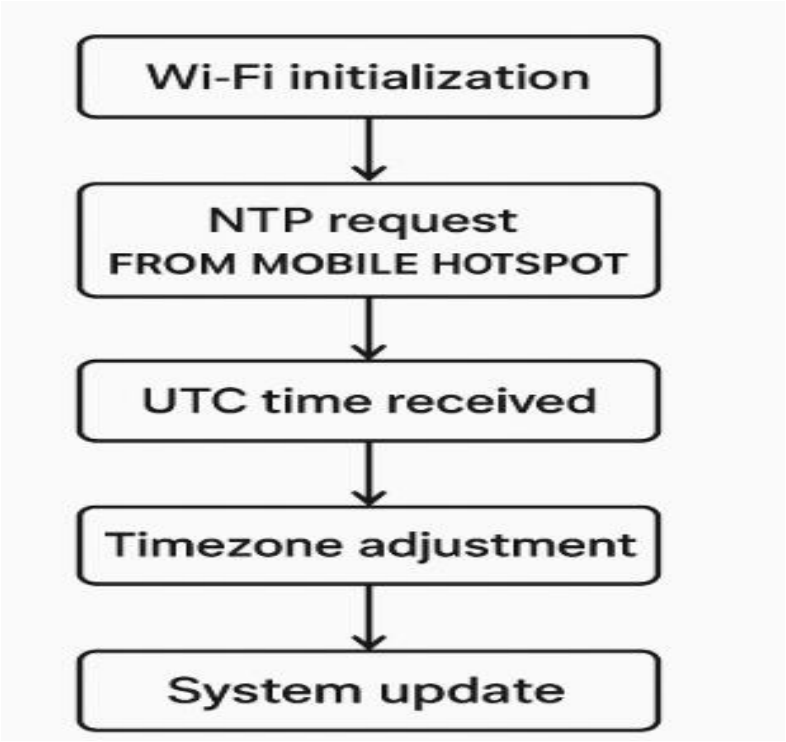


Figure 7-1:NTP synchronization workflow

7.2 Displaying Date & Time on TFT

The TFT display on the parent bracelet is used not only for directional guidance but also for presenting key user information, including the current date and time. Once synchronized, the device formats the retrieved time into a human-readable string (e.g., HH:MM:SS DD/MM/YYYY) and updates it on screen in real-time. This information is placed in a non-intrusive section of the circular screen layout, ensuring that it complements the directional arrow rather than distracting from it. The clock feature helps users stay oriented and can serve as a useful tool during timed monitoring scenarios, such as school pickups or scheduled caregiving shifts.

Figure 7-2:Date&Time Display

7.3 Battery Percentage Calculation and Display Monitoring

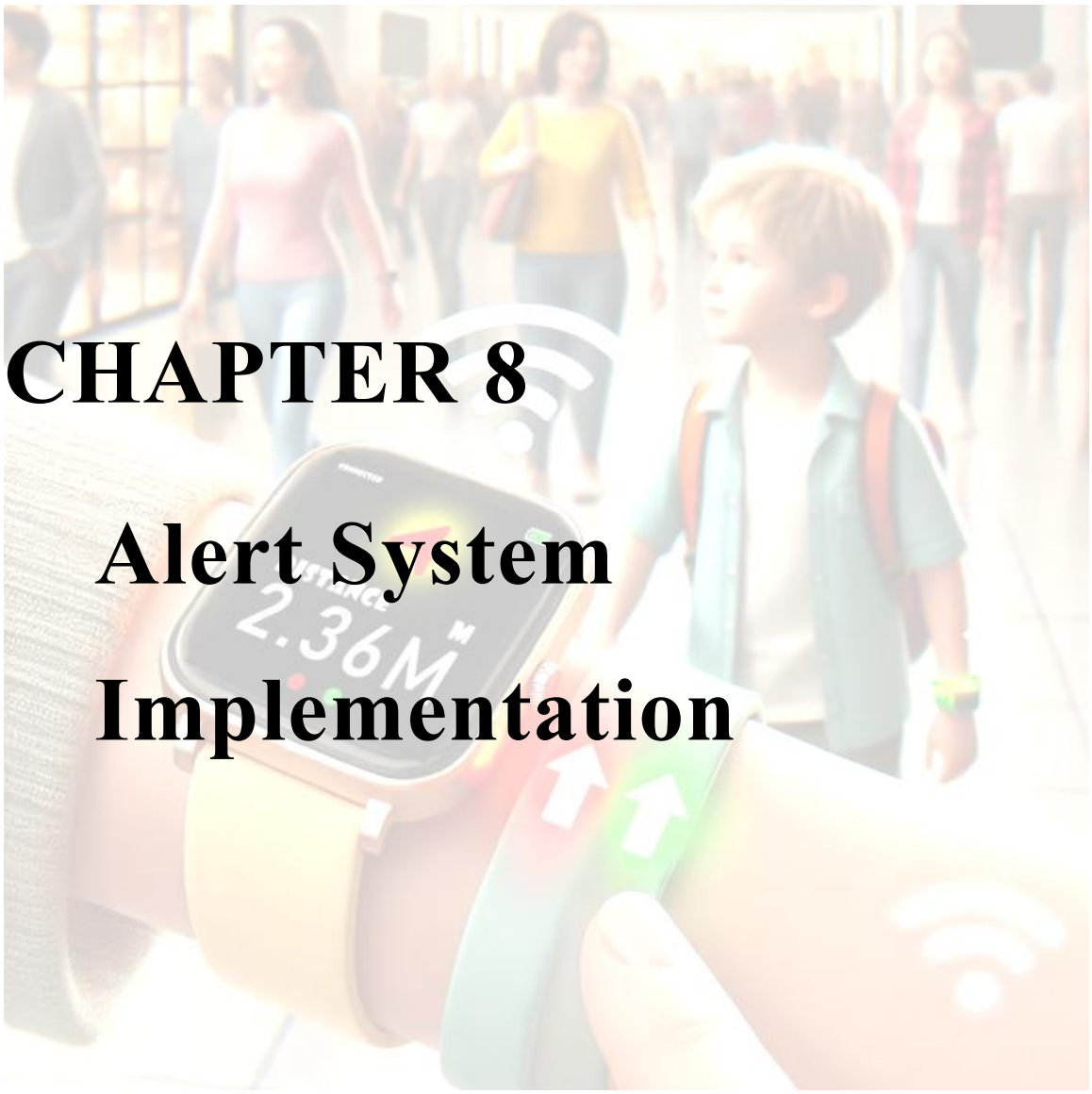
battery level is crucial for maintaining operational reliability. A voltage divider circuit is used to safely step down the Li-Po battery voltage and feed it into the ESP32's ADC pin. The software reads this analog value, converts it into a voltage measurement, and maps it to an estimated battery percentage based on a predefined calibration curve. The battery percentage is shown on the TFT screen as a numeric value and a graphical battery icon. The display updates in near-real-time and changes color when the battery is low, helping users take proactive action to recharge before the device powers down.

Figure 7-3:Battery display

7.4 Graphical Layout of the Screen

To make the TFT display both functional and user-friendly, the SafeLink user interface adopts a circular layout that takes full advantage of the GC9A01 round screen. The central part of the display is dedicated to the directional arrow, which rotates based on compass input. Around the arrow, additional UI elements are strategically placed: the time and date in the upper arc, battery percentage in the lower arc, and alert icons on the side regions. This layout maintains a clean and intuitive visual experience, allowing users to interpret multiple data points immediately without confusion or clutter. Icons are color-coded, and fonts are scaled for readability.

Figure 7-4: UI of TFT Screen



CHAPTER 8

Alert System Implementation

CHAPTER 8: Alert System Implementation

8.1 Triggering Buzzer and Vibration Based on Distance

One of the core safety features of SafeLink is its ability to provide real-time alerts through sound and vibration. These alerts are triggered based on the proximity zones determined by the RSSI-based distance estimation algorithm. When the RSSI value drops below predefined thresholds, the system classifies the child's distance as moderate or far. For moderate distance, only visual feedback may be given, but for far distances (critical separation), the system activates both the piezo buzzer and vibration motor on the parent bracelet. This dual-feedback mechanism ensures that the parent is immediately and unmistakably alerted, even in noisy or visually obstructed environments.

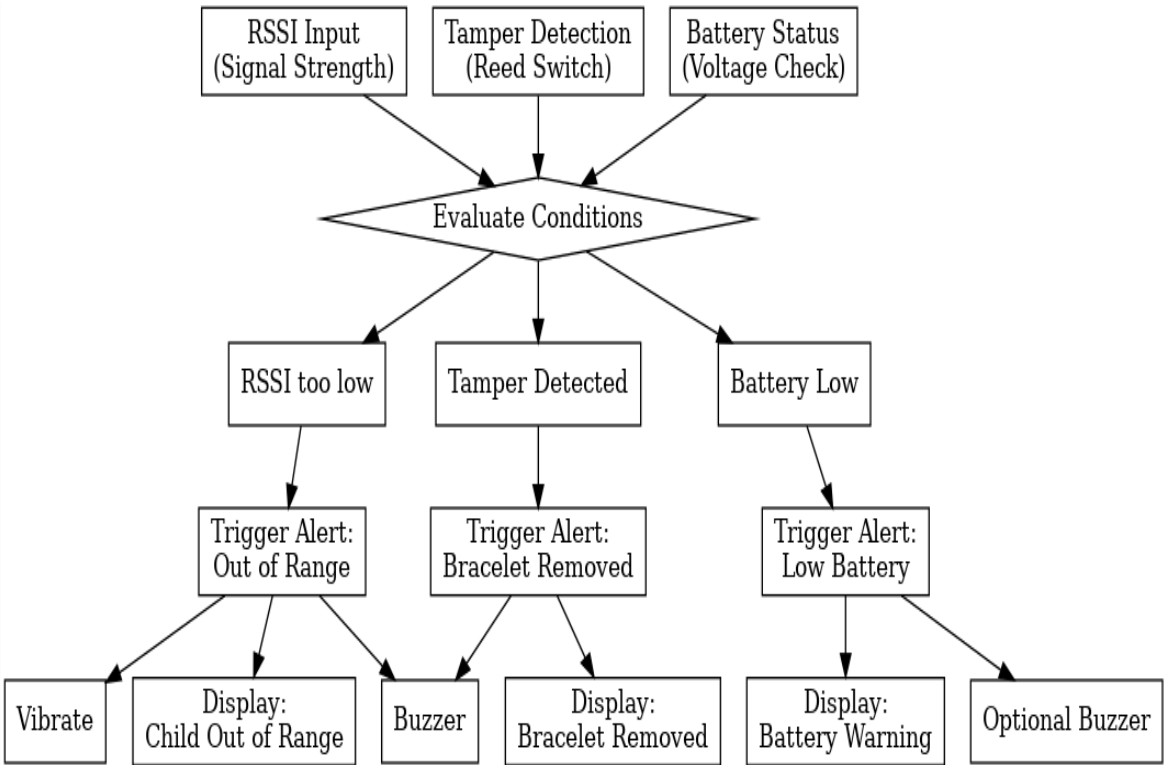


Figure 8-1: Alert System Mechanism

8.2 Tamper Detection via Reed Switch

The child bracelet includes a reed switch connected to a digital input pin on the ESP32. When the bracelet is properly worn, a magnet closes the switch. If the bracelet is removed or tampered with, the circuit opens, triggering an interrupt or polling-based check. This initiates an immediate alert on the parent bracelet, activating vibration, sound, and a red warning icon on the TFT display. The logic is lightweight and designed to respond instantly, ensuring the caregiver is notified in real-time if the bracelet is forcefully removed.

Figure 8-2:Tamper Detection Alert

8.3 Alert Handling Logic

The SafeLink alert system uses a modular logic framework to evaluate multiple triggers and activate the appropriate feedback channels. The core of this system is a decision tree:

- If RSSI < critical threshold → Activate buzzer, vibration, red arrow
- If RSSI in moderate range → Display yellow arrow only
- If tamper detected → Activate all alerts immediately
- If battery < critical voltage → Flash battery icon and vibrate once every minute This logic runs continuously in the main loop with interruptions available for immediate alert triggers. A priority system ensures that tamper and separation alerts override non-critical warnings.

8.4 Sound and Feedback Optimization

To ensure user comfort and maximize awareness, the buzzer and vibration patterns are carefully timed. Instead of continuous feedback, the buzzer uses a pulse pattern (e.g., 500 ms on, 500 ms off) to reduce annoyance and preserve battery. Similarly, the vibration motor is activated in short bursts. The sound frequency is selected to be within the most audible range for human ears (~2–4 kHz), and the vibration intensity is optimized to be noticeable but not uncomfortable when worn on the wrist. Future enhancements may include user-configurable alert settings through a companion app, or a configuration menu displayed on the TFT.

CHAPTER 9

PCB Design and Hardware Integration

CHAPTER 9: PCB Design and Hardware Integration

9.1 Overview of Parent and Child PCB

Both the parent and child bracelets rely on custom-designed PCBs (Printed Circuit Boards) to integrate all electronic components into a compact, wearable form

9.1.1 The parent PCB

Parent PCB accommodates the ESP32 microcontroller, TFT display header, compass module, buzzer, vibration motor, and battery monitoring circuits

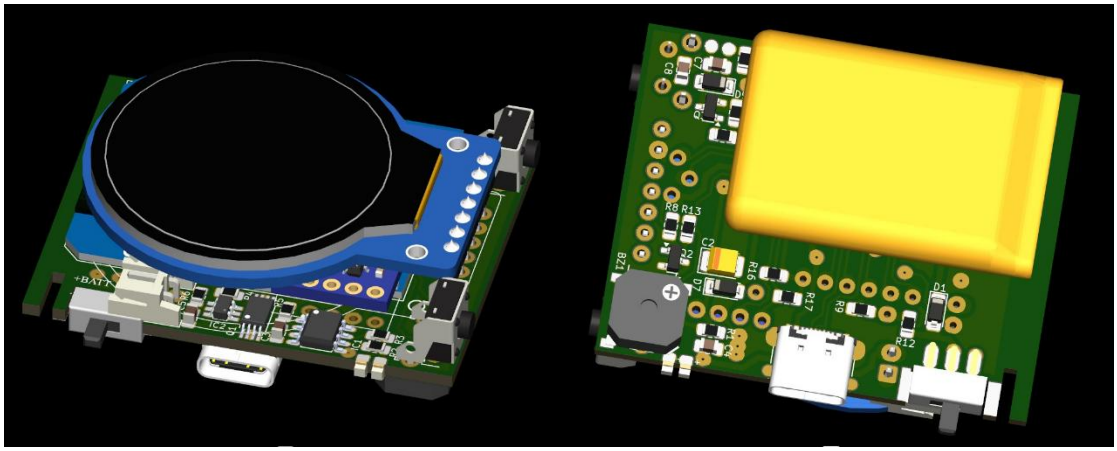


Figure 9-2: Parent Bracelet Top View

Figure 9-3 Parent Bracelet Bottom View

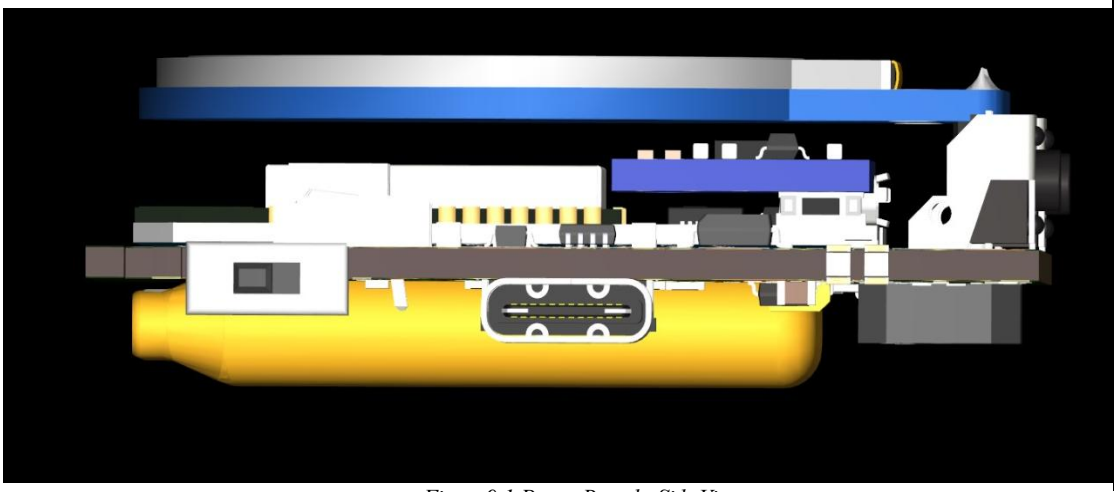


Figure 9-1:Parent Bracelet Side View

9.1.2 The child PCB

child PCB is smaller and simpler, containing the ESP32, reed switch interface, vibration motor driver, and power regulation.

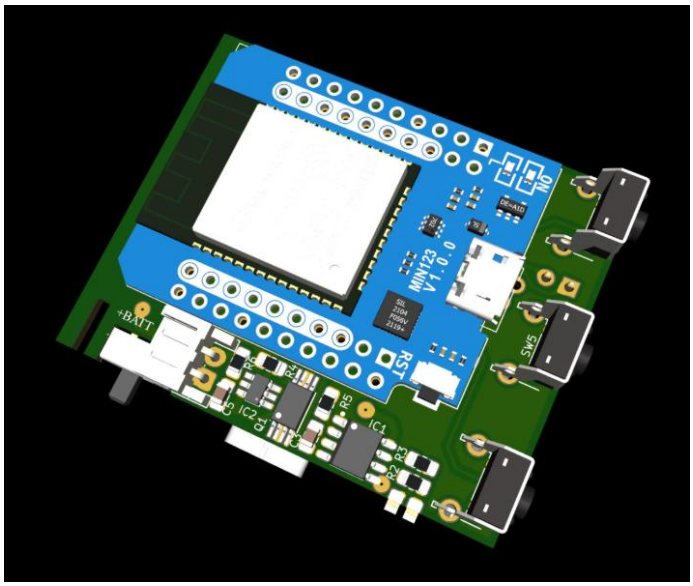


Figure 9-5:Child Bracelet Top View

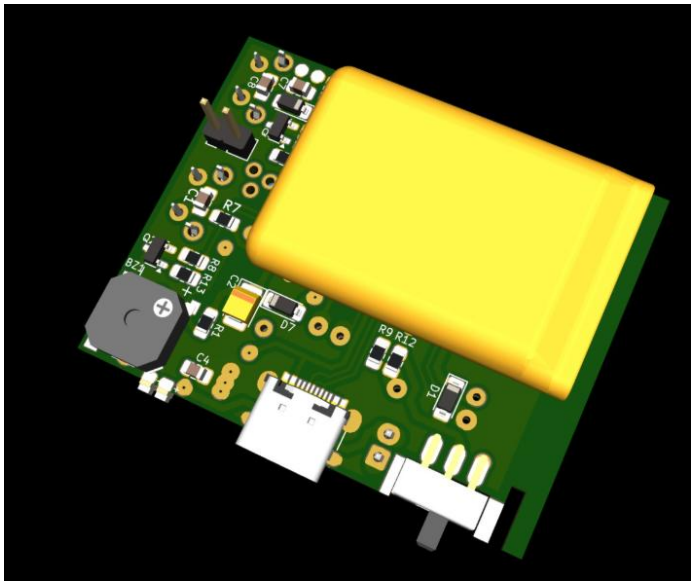


Figure 9-4:Child Bracelet Bottom View

Both designs prioritize minimal size, reliable connections, and efficient power layout to ensure long-term wearability and stable performance.

9.2 Circuit Schematic Explanation

Each bracelet has its own schematic layout developed in EDA (Electronic Design Automation) software such as KiCad. The schematic organizes components into logical blocks: power management, sensor interface, output devices, and communication modules. ESP32 is centrally positioned, with its GPIOs mapped to the respective peripherals. Protection components such as resistors, capacitors, and flyback diodes (for inductive loads like the vibration motor) are included to safeguard circuit integrity.

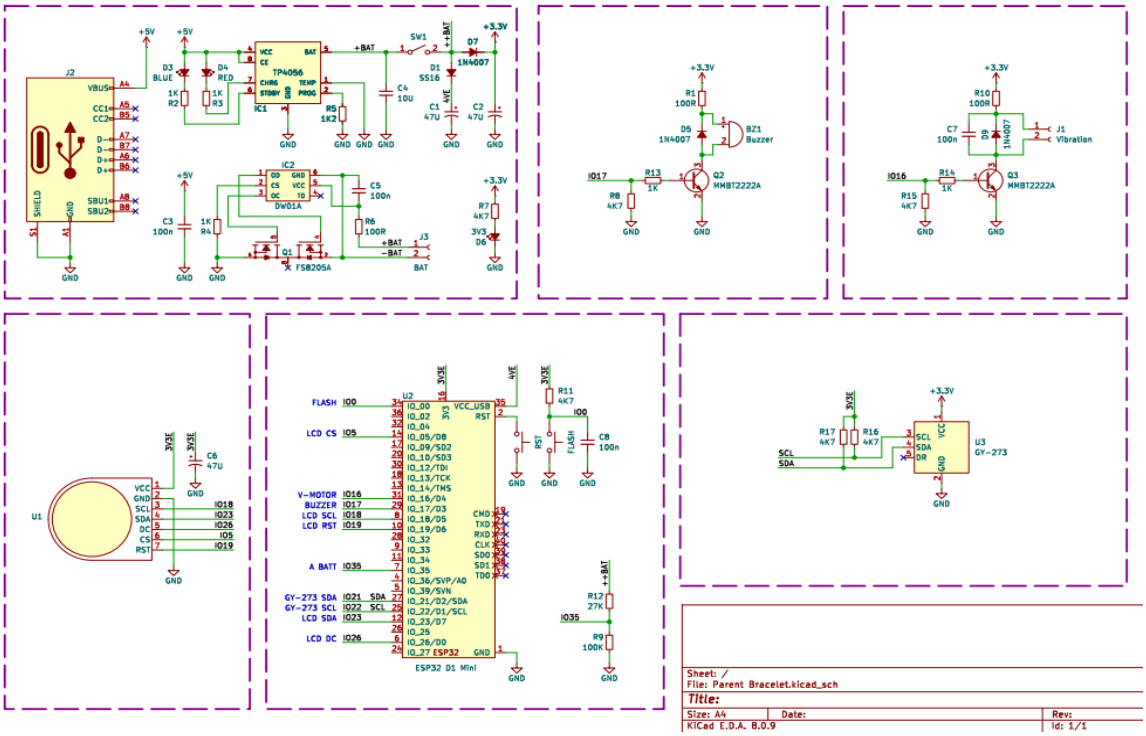


Figure 9-6:Parent Bracelet KiCad Schematic

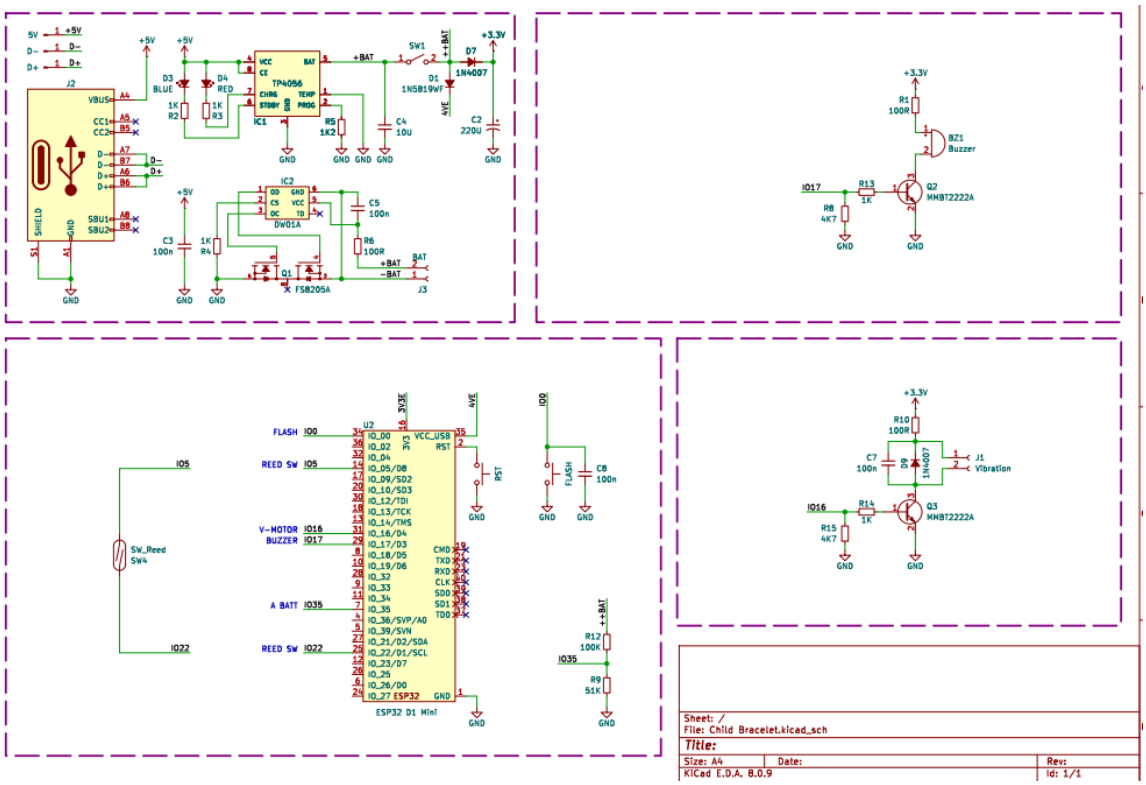


Figure 9-7:Child Bracelet KiCad Schematic

9.3 Pin Mapping and Component Layout For clarity and modularity

pin mapping is consistent across both bracelets. Digital GPIOs are used for reed switch and vibration motor control, I2C lines connect to the compass and potential expansion sensors, and analog pins handle battery voltage sensing. On the physical PCB layout, components are grouped by function: power input and charging on one side, processing core at the center, and output modules near the bracelet edge for easier connection to the housing. Ground planes are used to reduce electrical noise and enhance stability.

9.4 Design Considerations (Space, Heat, Orientation)

Designing a wearable PCB requires attention to multiple factors:

- **Space:** All components must fit within the bracelet's casing, leaving room for wiring and connectors.
- **Heat:** Power regulators and the ESP32 can produce heat; components are spaced to prevent hotspots.
- **Orientation:** Components like the compass must be oriented properly to ensure accurate reading. The PCB layout accounts for rotational alignment relative to the user's wrist.
- **Accessibility:** USB charging ports and reset buttons are placed at the edge for user access.

9.5 Power Distribution and Trace Widths

The power distribution network is designed to minimize voltage drops and overheating. Thicker copper traces are used for power lines, especially those powering the vibration motor and display. Ground and VCC rails are routed with sufficient width (typically 1–2 mm) and short paths to reduce resistance. Capacitor banks near the ESP32 stabilize voltage during sudden load changes.

9.6 Final PCB Dimensions and 3D View

The final parent PCB has dimensions of approximately 38mm × 38mm in a circular shape to match the GC9A01 display and round casing. The child PCB is a compact rectangular layout (around 28mm × 20mm) to fit snugly within its enclosure. 3D renders generated in KiCad were used to verify spacing, component alignment, and mounting hole positions. These visual models also assisted in casing design and wire harness planning.

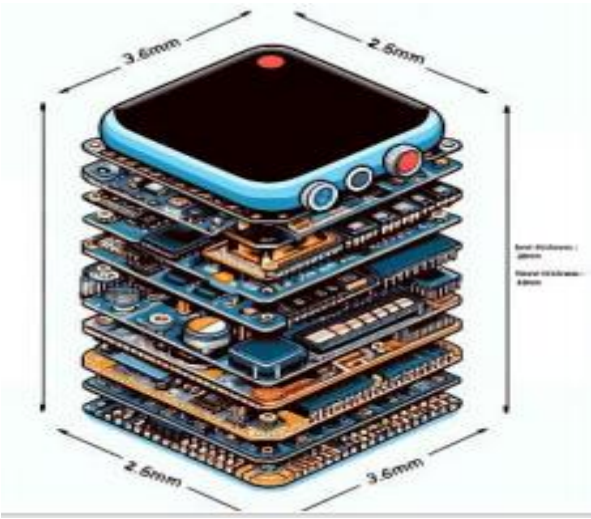
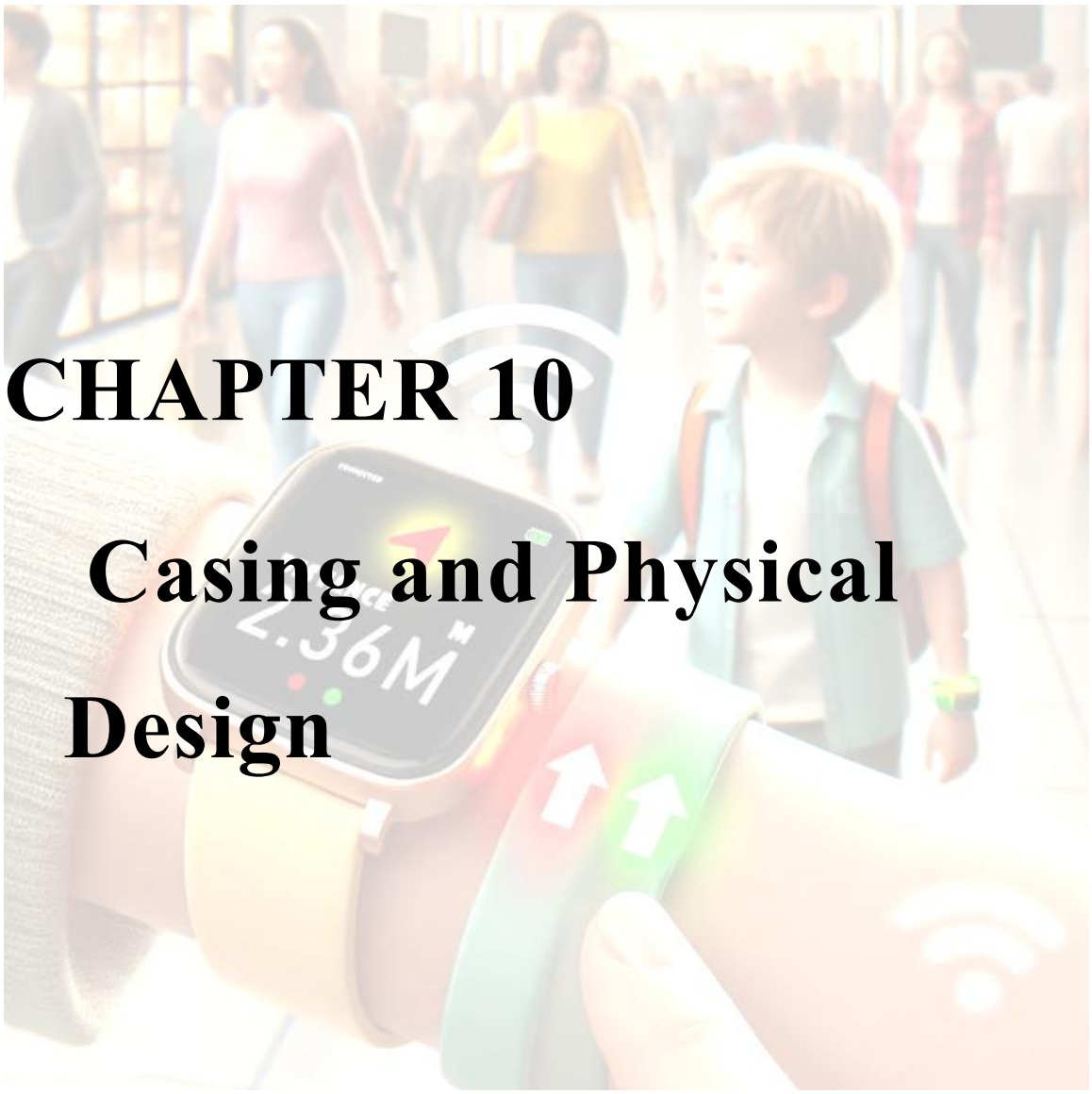


Figure 9-8:Exploded View of Smartwatch PCB Stackup



CHAPTER 10

Casing and Physical Design

CHAPTER 10: Casing and Physical Design

10.1 3D Watch Casing Design Tools

The SafeLink bracelet casing was designed using CAD software such as SolidWorks, enabling precise modeling of the round form factor required by the GC9A01 TFT display. These tools facilitated designing snap-fit housing, ensuring easy assembly and maintenance while keeping the bracelet compact and ergonomic.

Figure 10-1:Parent Bracelet 3D Design Prototype

10.2 Internal Layout and Mounting Strategy

The casing design considered optimal placement of the PCB, battery, and wiring harnesses to minimize vibrations and ensure reliable compass operation. Internal standoffs were added to secure the PCB while avoiding contact with the casing walls, reducing magnetic interference. Cutouts were precisely aligned with the USB port for convenient charging access.

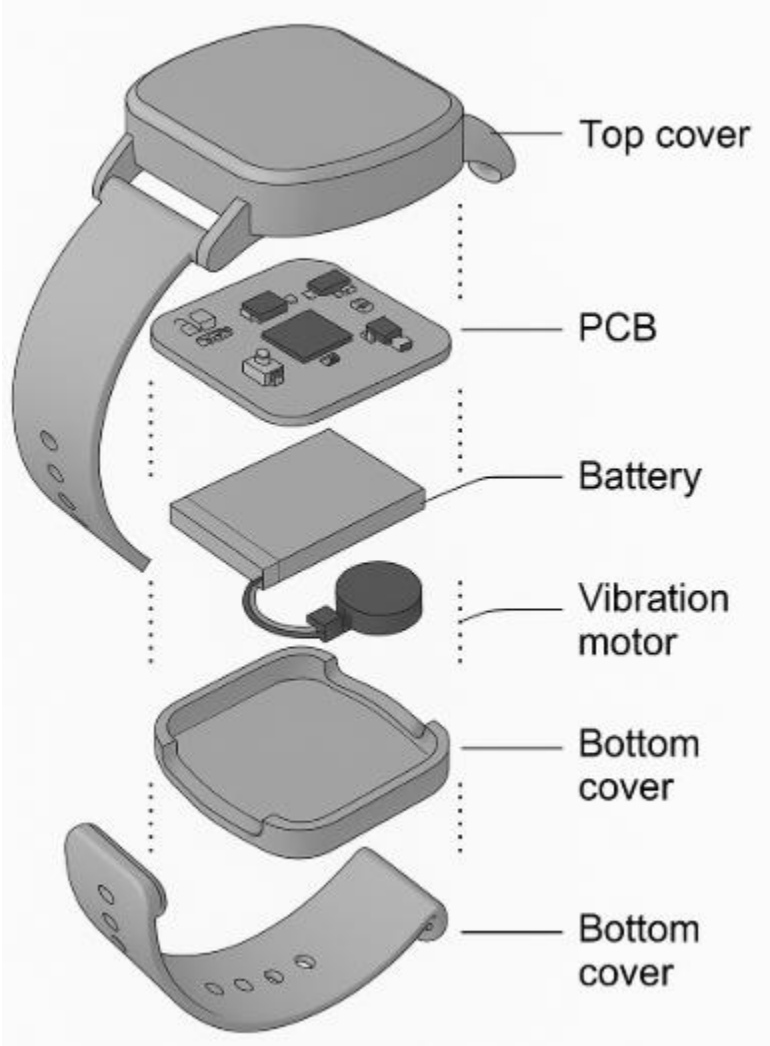


Figure 10-2:Exploded view of bracelet assembly

10.3 Material Selection and Durability

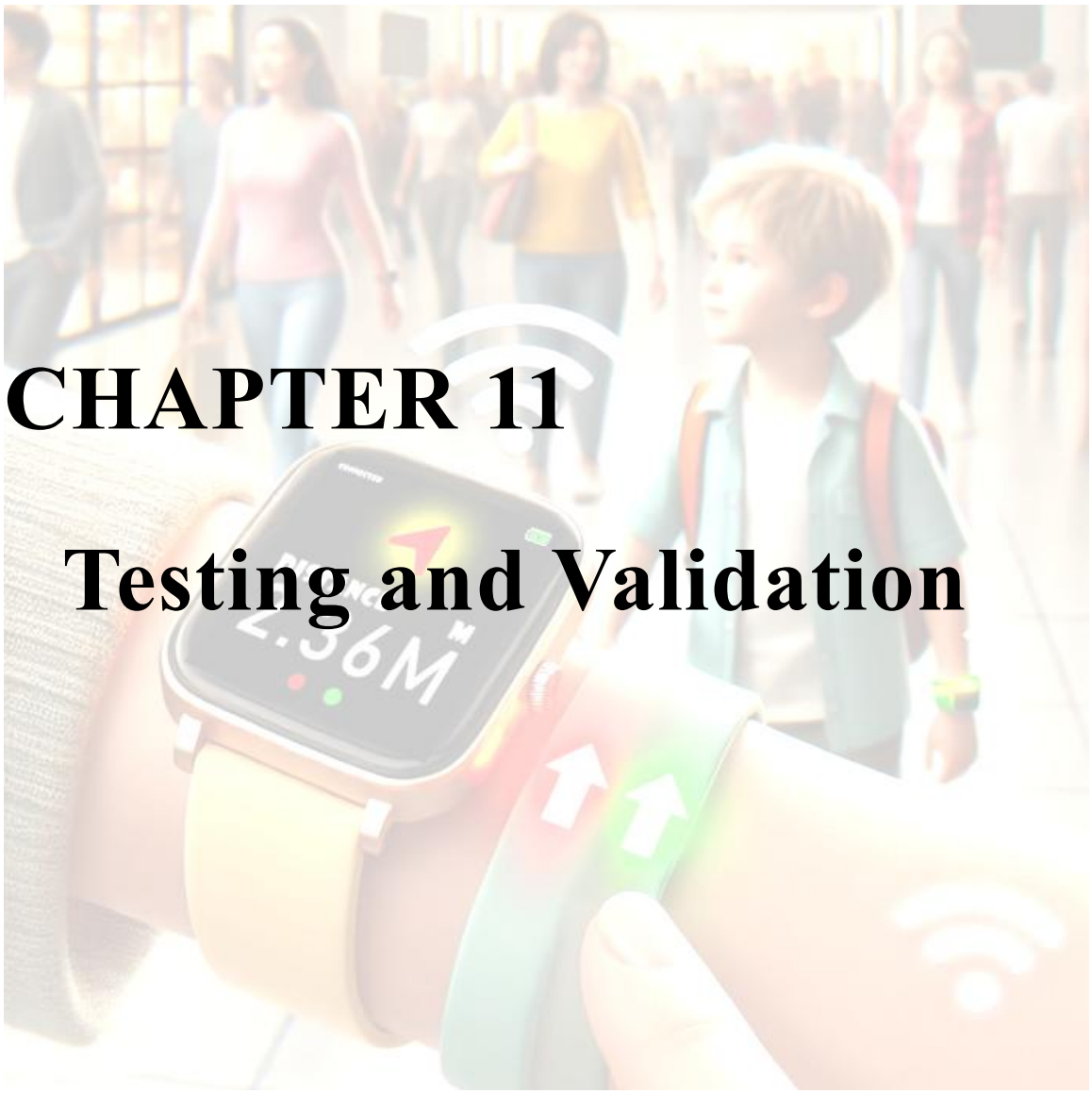
For prototypes, PLA and PETG filaments were chosen for 3D printing due to their balance of strength, printability, and wear resistance. Future iterations will explore injection-molded ABS or polycarbonate for better durability, waterproofing, and a professional finish suitable for everyday wear.

Figure 10-4:PLA Filaments

Figure 10-3:PETG Filaments

10.4 Prototypes and Enclosure Challenges

Several prototype iterations were printed and assembled, revealing challenges such as insufficient space for wiring and unexpected interferences between components. These issues led to redesigning improving internal clearance and simplifying the assembly process. User feedback during pilot testing highlighted preferences for lighter weight and softer materials for improved comfort.



CHAPTER 11

Testing and Validation

CHAPTER 11: Testing and Validation

11.1 Functional Testing

Each module within the SafeLink system underwent dedicated functional testing to ensure operational integrity. The ESP32 boards were first tested for boot reliability, pin configuration, and basic serial communication. Peripherals such as the compass, display, buzzer, and vibration motor were then individually verified using test sketches in the Arduino IDE. For example, the compass sensor was validated by comparing raw readings with an external magnetic compass to confirm directional consistency. Similarly, reed switch logic was tested with a simulated magnetic field to trigger tamper alerts, and battery monitoring was verified across different voltage levels. Only after passing module-level tests did the components proceed to system integration.

11.2 Distance Estimation Accuracy

RSSI-based distance estimation was tested in both controlled indoor environments and real-world scenarios such as open areas and crowded spaces. Tests involved placing the child bracelet at known distances from the parent device and recording RSSI values over time. These values were then averaged and mapped to predefined distance zones (Close, Moderate, Far). Results showed that while RSSI is inherently variable due to environmental noise and interference, the zone-based feedback system (with hysteresis) effectively reduced erratic behavior. Accuracy was highest in open-indoor areas, with consistent zone transitions observed.

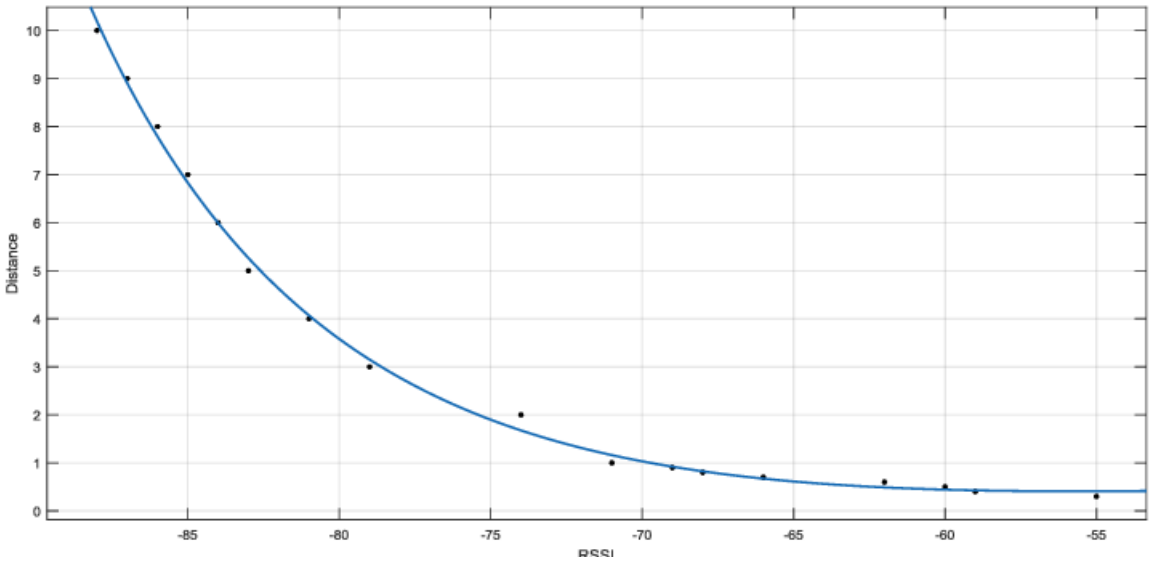


Figure 11-1:Distance vs RSSI curve plot

11.3 Compass Calibration Validation

To ensure compass reliability, the parent bracelet was rotated in various orientations during initialization. Raw magnetic readings were plotted to observe distortion patterns, and a hard-iron calibration algorithm was applied to center the magnetic vector. Calibration success was validated by comparing the computed heading with known directional references. After calibration, the heading error was reduced to within ±5° in stable conditions. Recalibration was recommended after battery replacement or major physical repositioning.

11.4 Battery Lifetime Estimation

Battery performance was tested by running the bracelets under realistic use conditions, including standby, active tracking, and alert-triggering states. The parent device (with display and alerts) lasted between 5 to 7 hours under moderate usage with intermittent alerts. The child device, operating in broadcast-only mode with minimal feedback, lasted

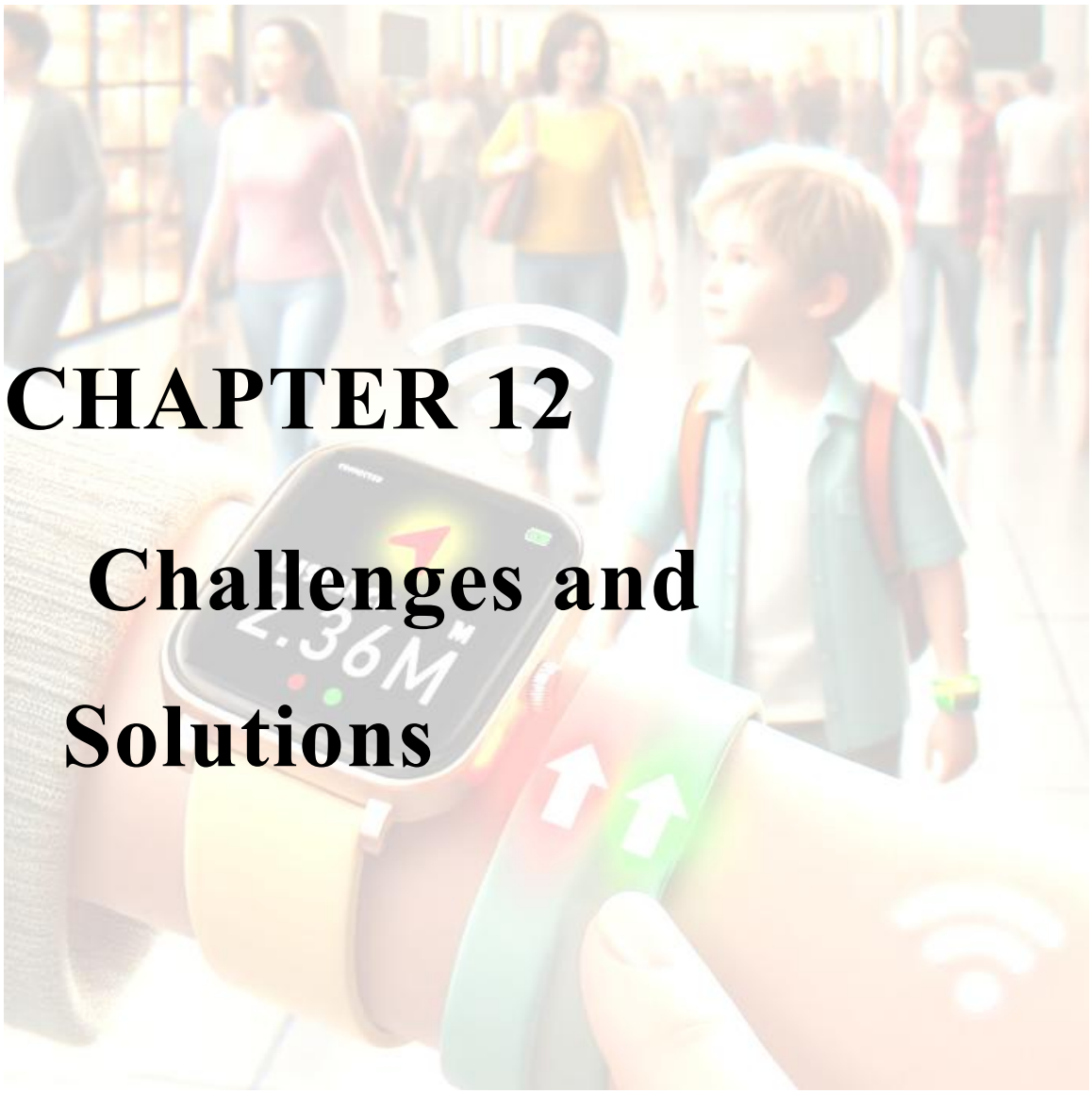
approximately 9 to 12 hours. Battery drain graphs were recorded to determine energy consumption rates, and future optimization (e.g., sleep modes) was identified as a key area for improvement.

11.5 Troubleshooting and Fixes

During the testing phase, several issues were identified and addressed:

- **Issue:** TFT screen would occasionally freeze during compass updates. **Fix:** Added delay between screen and I2C updates to prevent bus conflict.
- **Issue:** Wi-Fi disconnection after prolonged idle time. **Fix:** Implemented reconnection loop with heartbeat check.
- **Issue:** Vibration motor drained battery during frequent alerts. **Fix:** Optimized alert logic with cooldown timer.
- **Issue:** False tamper triggers due to weak magnet. **Fix:** Replaced magnet with a neodymium version for stronger hold.

This iterative process of identifying faults and applying real-time fixes helped significantly stabilize the system for final deployment.



CHAPTER 12

Challenges and Solutions

CHAPTER 12: Challenges and Solutions

12.1 RSSI Accuracy Issues

One of the primary challenges encountered during development was the inconsistency of RSSI readings, which are sensitive to environmental factors such as walls, reflective surfaces, and electromagnetic interference. Variations caused by human movement or obstacles frequently lead to false proximity alerts. To address this, a hysteresis-based algorithm was implemented, requiring several consecutive RSSI samples outside a threshold before triggering an alert. This reduced false positives and improved reliability without sacrificing responsiveness.[16]

12.2 Compass Inaccuracy or Fluctuation

The compass sensor (QMC5883L) is highly susceptible to magnetic interference from nearby metal objects or electronic devices. During testing, large heading errors were observed when the bracelet was used near cars, elevators, or steel structures. To mitigate this, a calibration process involving full 360° rotation was required during setup, combined with real-time filtering using moving averages. A recommendation was added to avoid wearing the bracelet near strong magnetic fields.

12.3 GPS Module Abandonment

The project planned to integrate a GPS module for extended-range outdoor tracking. However, testing revealed several limitations: high power consumption, poor indoor signal reception, and increased complexity in both software and PCB design. Additionally, GPS-based solutions were outside the intended local-use scope of SafeLink. As a result, the GPS module was excluded, and the focus shifted to optimizing short-range Wi-Fi RSSI and compass-based guidance, which better aligned with indoor and localized outdoor applications.[25]

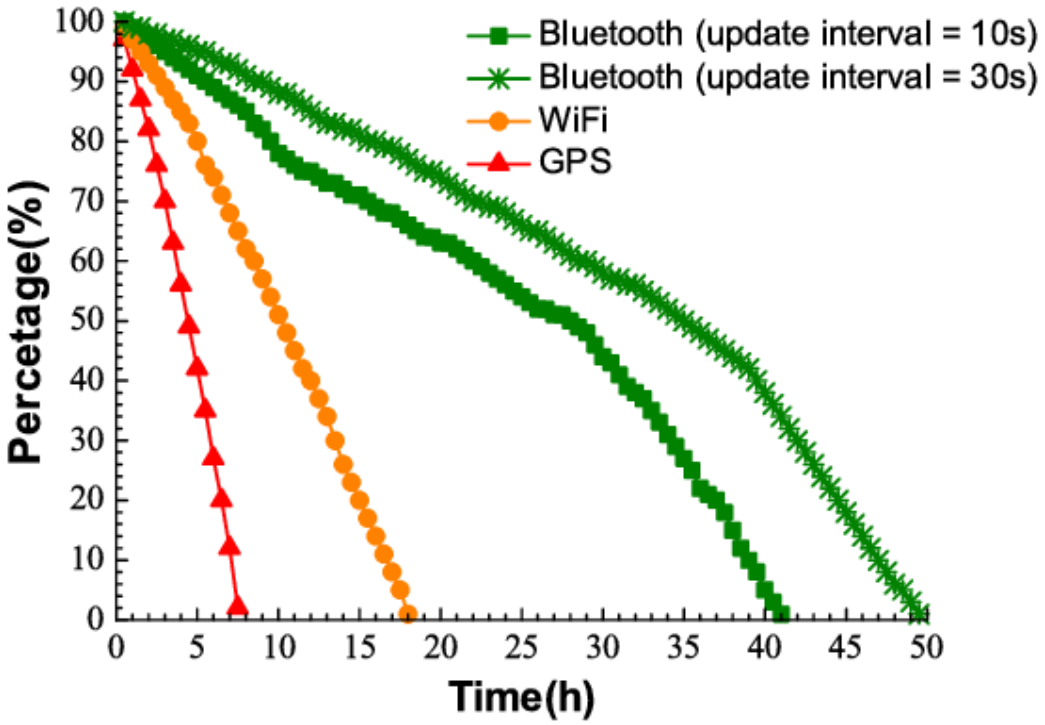


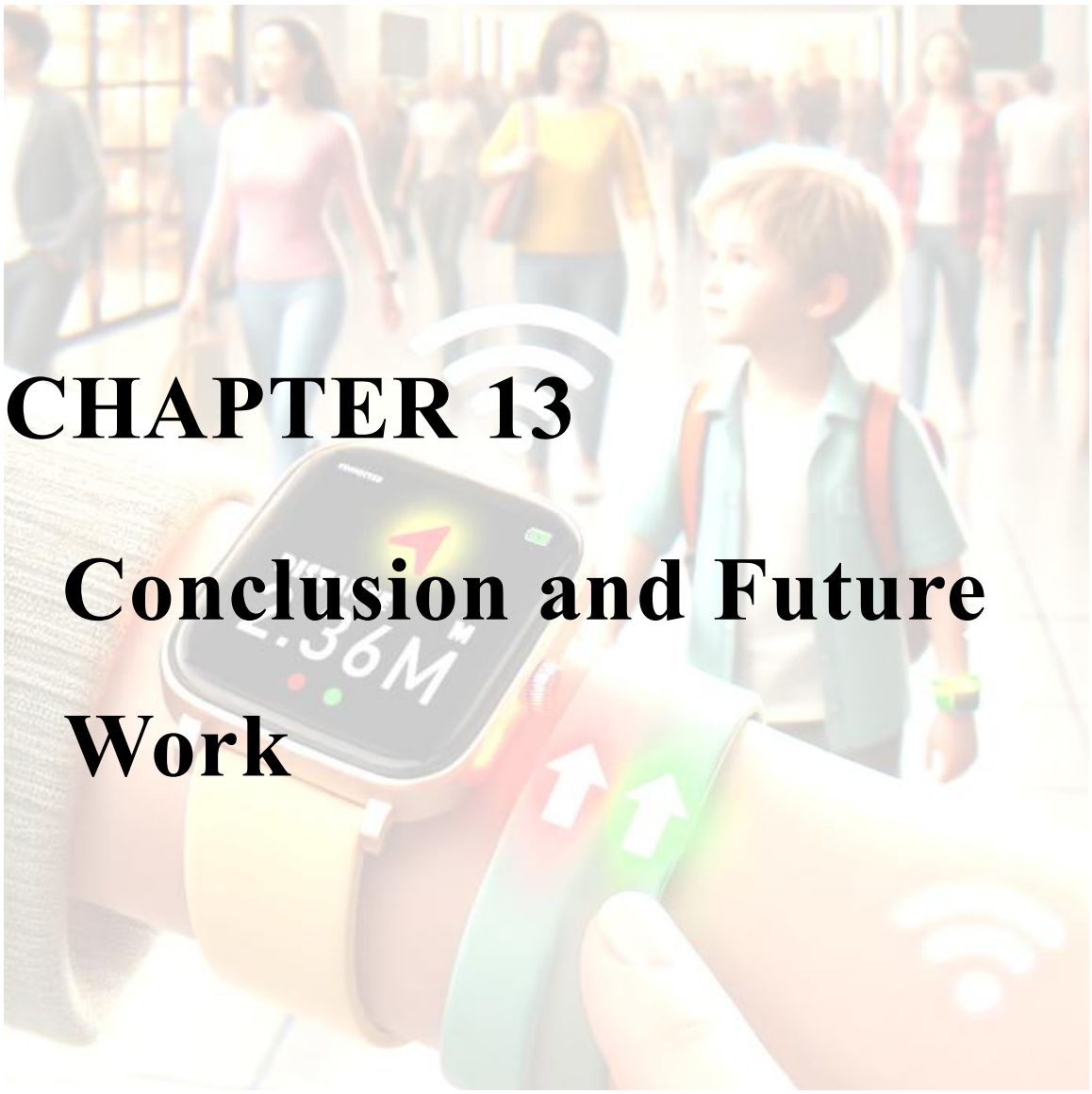
Figure 12-1:Energy Consumption of GPS & WiFi

12.4 PCB Fabrication Constraints

Designing compact, wearable PCBs introduced significant manufacturing challenges. Early prototypes suffered from misaligned vias, solder bridges, and insufficient clearance between traces and pads, leading to unreliable operation. Solutions included refining design rules within the PCB software, increasing critical trace widths, adding protective silkscreen markings, and collaborating closely with PCB manufacturers to review design for manufacturability (DFM) guidelines. These improvements enhanced yield and reduced rework time.

12.5 Display or Power Failure

Handling Several prototypes experienced unexpected TFT display freezes or complete power drops, often traced back to voltage dips when the vibration motor or buzzer activated simultaneously with display refresh cycles. A capacitor bank was added near the power input to buffer transient loads, and software logic was modified to stagger high-current events. This approach stabilized system behavior and reduced occurrences of total device resets during alerts.



CHAPTER 13

Conclusion and Future Work

CHAPTER 13: Conclusion and Future Work

13.1 Summary of Achievements

The SafeLink project successfully delivered a wearable system designed to enhance safety for children and elderly individuals through real-time proximity alerts and directional guidance. The system achieved reliable short-range communication using Wi-Fi RSSI, intuitive visual feedback with a compass-based arrow, and effective alert mechanisms through vibration, sound, and visual cues. Through careful hardware integration, modular firmware development, and iterative testing, SafeLink met its core objectives of providing a low-cost, power-efficient, and easy-to-use solution suitable for parents, caregivers, and institutional settings.

Figure 13-1:SafeLink Highlights

13.2 Benefits to Society and Target Users

SafeLink offers immediate social value by empowering parents and caregivers to respond quickly to separation events, reducing the risk of child abduction or elderly wandering. By providing clear directional guidance and instant alerts, it alleviates stress, enhances confidence, and promotes independence for vulnerable users. Institutions such as daycare centers and senior living facilities can also deploy SafeLink to improve supervision standards and reduce staff workload, contributing to safer communities overall.

13.3 Proposed Future Enhancements

13.3.1 GPS Re-Integration

Future versions could include an optimized, low-power GPS module for optional outdoor tracking, enabling parents to locate children beyond Wi-Fi range without compromising battery life through advanced power management.[7]

13.3.2 Mobile App Companion

A smartphone application could offer remote notifications, configuration options, and historical movement logs, providing additional convenience and peace of mind for caregivers.

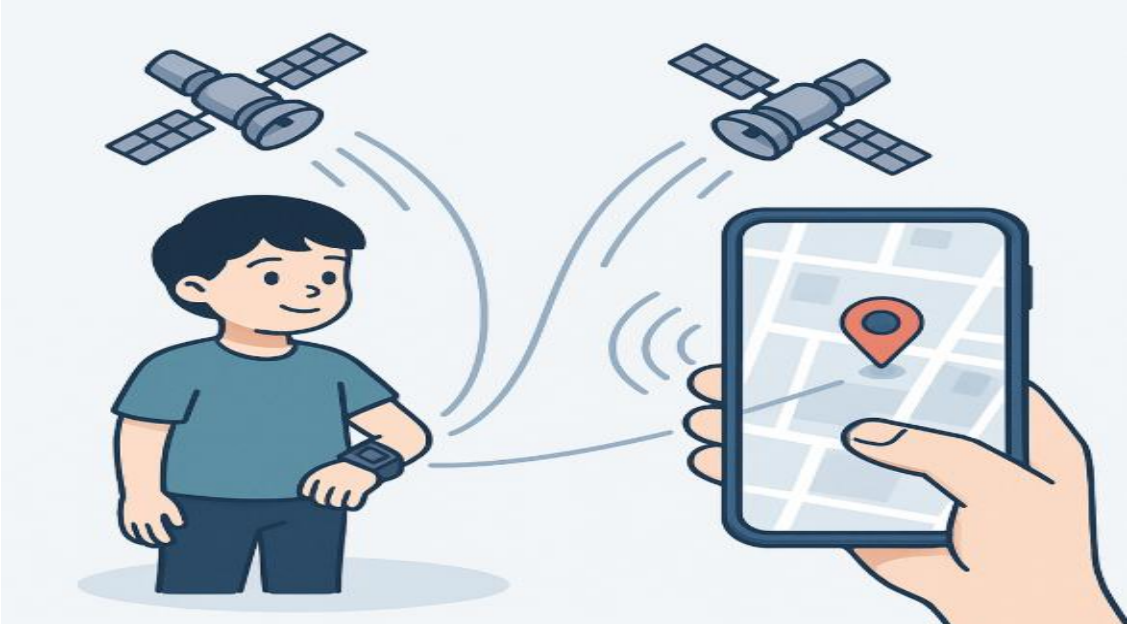


Figure 13-2:GPS integration & Mobile App

13.3.3 SOS Button and Emergency Call

An SOS button integrated into the child bracelet could trigger emergency alerts to the parent's bracelet or connected smartphone, adding an extra layer of protection during emergencies.

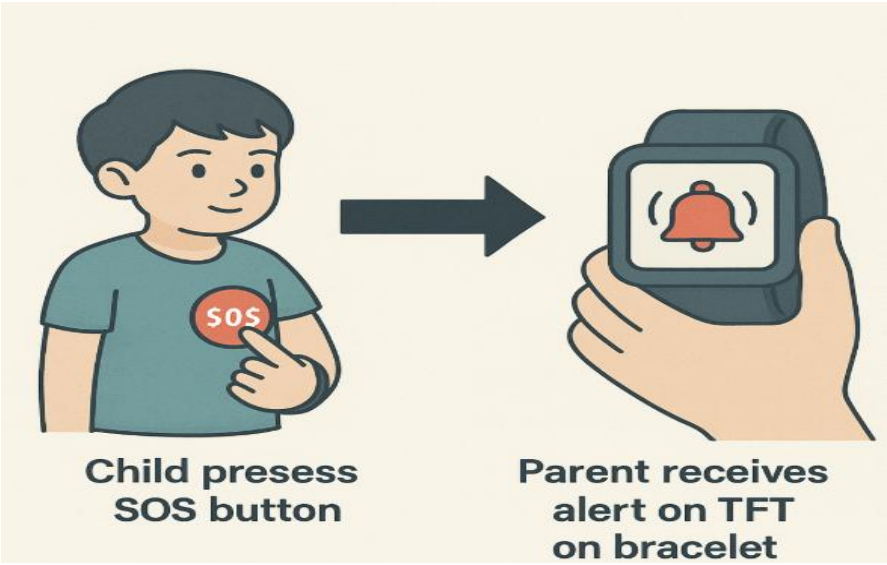


Figure 13-4:SOS working mechanism

Figure 13-3:SOS figure

13.3.4 Ultra-Wideband (UWB) for Precise Location Integrating

Ultra-Wideband technology would provide centimeter-level distance and direction measurement, overcoming RSSI's limitations in accuracy and environmental sensitivity.[10]

	UWB	BLE	Wi-Fi	GPS	Camera	LIDAR	Passive RFID
Positioning Accuracy	10-30 cm	5 meter	5-15 meter	5-10 meter outdoor only	10-30 cm	1 cm	not applicable
Scalability	✓	✓	✓	✓	✗	✗	✓
Real-time	✓	✓	~	~	✓	✓	✗
No dedicated infrastructure	✗	✗*	✗ ✓	✓	✓	✓	✗
Range	20-30 m	20 m	30 m	not applicable	20 m	60 - 100 m	5cm - 5m
Tracker cost	low	very low	medium	medium	very high	very high	ultra low

Figure 13-5:UWB vs WiFi

13.3.5 Indoor Navigation

Future iterations could include indoor positioning using BLE beacons or Wi-Fi fingerprinting, helping caregivers locate children or elderly individuals within large buildings such as malls, hospitals, or schools.

The SafeLink platform's flexible architecture ensures it can evolve to meet these enhancements, positioning it as a powerful tool in the growing field of personal safety technology.

CHAPTER 14

Appendices

CHAPTER 14: Appendices

14.1 Datasheets of Components

This appendix contains datasheets or summarized specifications for the major electronic components used in the SafeLink system. These documents provide essential technical details such as operating voltage, current consumption, pin configurations, communication protocols, and recommended usage guidelines. The components included are:

- D1 Mini ESP32

  https://www.espressif.com/sites/default/files/documentation/esp32-wroom-32_datasheet_en.pdf

- QMC5883L

  https://www.farnell.com/datasheets/1683374.pdf

- GC9A01 TFT Display

  https://www.buydisplay.com/download/manual/ER-TFTM1.28-1_Datasheet.pdf

- KLJ-8530-3627 Piezo Buzzer

  https://lcsc.com/datasheet/lcsc_datasheet_2410010300_KELIKING-KLJ-8530-3627_C189206.pdf

- Coin Vibration Motor

  https://nfpshop.com/wp-content/uploads/2022/05/SPEC-NFP-FLAT-C0820.pdf

- Reed Switch

  https://standexelectronics.com/wpcontent/uploads/OKI_Reed_Switch_ORD213.pdf

- TP4056 Charging Module

  https://components101.com/sites/default/files/component_datasheet/TP4056A%20-%20Li-ion%20Battery%20Charging%20Module_0.pdf

- Li-Po Battery 3.7V

  https://www.farnell.com/datasheets/2369105.pdf

14.2 User Guide

This appendix provides a simplified quick start guide for SafeLink users, ensuring ease of operation and a positive user experience.

14.2.1 Device Setup

- Charge both parent and child bracelets fully before first use (using TP4056 modules).
- Switch on the parent bracelet; ensure the welcome screen appears.
- Power on the child bracelet; pairing completes automatically within a few seconds.

14.2.2 Operation

- Wear the bracelets securely using the adjustable bands.
- Monitor the TFT screen: an arrow indicates the direction
- The visual color of the arrow changes based on distance:
- Green → Close
- Yellow → Moderate
- Red → Out of Range
- The vibration motor and buzzer activate if separation exceeds safety thresholds.

14.2.3 Tamper Detection

- The child bracelet's reed switch triggers an alert if someone attempts to remove or open the bracelet.
- Parent's bracelet shows a tamper alert icon and activates vibration/buzzer.

14.2.4 Battery Status

- Battery level is displayed on the parent TFT as a percentage.
- Charge devices when battery drops below 20% to avoid unexpected shutdowns.

14.3 Cost Estimation & Bill of Materials (BOM)

This appendix details the estimated cost for producing a single set of SafeLink bracelets, including quantities, per-unit prices, and total cost estimates. Prices reflect typical online component costs as of the time of development. Final costs may vary depending on supplier agreements, shipping, and production scale.

Component Name	Description	Amount	COST
D1 mini ESP32	WiFi+Bluetooth development board	2	750 LE
GC9A01 TFT LCD Display (Round)	1.28" or 1.3" circular TFT screen	1	680 LE
QMC5883L Magnetometer	Digital 3-axis compass sensor	1	250 LE
SMD Buzzer (KLJ-8530-3627)	Sound alert component	2	90 LE
Vibration Motor	Provides tactile feedback	2	30 LE
Reed Switch	detect proximity of a magnet	1	15 LE
Li-Po Battery (3.7V, 500–1000mAh)	Rechargeable lithium polymer battery	2	280 LE
TP4056 Charging Module	A small module used to safely charge Li-Po batteries via USB	2	70 LE
AMS1117	Voltage Regulator 3.3V	2	8 LE

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Research & Journals

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[2]Şeker, Ö., Akdoğan, T., Şahin, B., and Dalkılıç, G. Physical Tracking of ESP32 IoT Devices with RSSI Based Indoor Position Calculation. Journal of Millimeter Wave Communication, Optimization and Modelling, vol. 4(1), 2024.

[3]Amritha, T.J., and Rukmini, R. Child Monitoring System Using Smart Wearable Device with Proximity Sensor and Cloud Integration. Materials Today: Proceedings, vol. 62 Pt. 5, 2022.

[4]https://pubmed.ncbi.nlm.nih.gov/31067819/

[5]https://link.springer.com/article/10.1007/s11042-018-7134-7

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[7]https://arxiv.org/abs/2107.09509

[8]https://link.springer.com/chapter/10.1007/978-3-030-42934-8_15

[9]https://www.mdpi.com/1424-8220/17/2/341

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[13]https://arxiv.org/abs/1912.07801

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[15]https://www.researchgate.net/publication/316176383_Child_safety_wearable_device

[16]https://www.researchgate.net/publication/375957176_Studies_on_indoor_positioning_algorithms_using_BLE_Beacons

Patents

[17]https://patents.google.com/patent/US6847295B1/en

[18]https://patents.google.com/patent/WO2017185722A1/en

DIY, Open Source & Tutorials

[19]https://www.electroniclinic.com/how-to-make-smart-bracelet-for-long-distance-couples-using-esp32/

[20]https://www.instructables.com/ESP32-Based-Wireless-Smartwatch-LiDAR-Distance-Sen/

[21]https://github.com/muhammadghassan/RSSI-Based-Indoor-Localization-using-ESP32

[22]https://github.com/SlimmShadyyy/BLE-Proximity-Tracker-RSSI-Based-Distance-Estimation-using-ESP32

[23]https://github.com/comfy-couch/ESP32-GPS-Wifi-Mapper

[24]https://github.com/keepworking/Mecha_QMC5883L

[25]https://forum.arduino.cc/t/hmc5883l-sensor-problem/549749

Community Discussions & Technical Help

[26]https://www.reddit.com/r/Autism_Parenting/comments/13vvtpe/tracking_devices_for_children_with_sensory_and/

[27]https://www.reddit.com/r/arduino/comments/16x1sjl/proximity_detection_and_distance_measurement_of/

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Datasheets & Hardware Resources

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